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Chemistry
Class 11

Sindh Textbook Board

(English Medium)

Based on the Sindh Textbook Board Book

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Stoichiometry

Unit 2
Atomic Structure

Unit 3
Theories of Covalent Bonding and Shapes of Molecules

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State of Matter I: Gas

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Stoichiometry

UNIT 1

Mole and Avogadro's Number

LESSON 1

Stoichiometry is the study of the quantitative relationship between reactants and products in a chemical reaction using a balanced chemical equation. The central counting unit of stoichiometry is the mole, which links the mass of a substance with the number of its particles.

1. The Mole

A mole is the gram atomic mass, gram molecular mass or gram formula mass of a substance, and it contains 6.02 x 10^23 particles (atoms, molecules or ions). For example, one mole of carbon atoms weighs 12g and contains 6.02 x 10^23 atoms, while one mole of water weighs 18g and contains 6.02 x 10^23 molecules.

2. Avogadro's Number

Avogadro's number is the number of particles present in one mole of any substance, equal to 6.02 x 10^23 and denoted by NA. It is a constant; it does not change with temperature. With its help the mole becomes a counting unit like a dozen: just as one dozen means 12 objects, one mole means 6.02 x 10^23 particles. The mole and Avogadro's number are interrelated because the mole is defined through it: one mole of every substance, whatever its mass in grams, always contains Avogadro's number of particles.

Number of moles = Number of particles / 6.02 x 10^23

3. Molar Volume

Molar volume is the volume occupied by one mole of any gas at STP (273K and 1 atm pressure); for all ideal gases its value is 22.4 dm3. So 44.8 dm3 of O2 at STP = 44.8/22.4 = 2 moles = 2 x 6.02 x 10^23 = 12.04 x 10^23 molecules.

4. Worked Numericals

Numerical 1: Calculate the number of moles and molecules in (i) 38g of carbon disulphide (CS2) (ii) 68.4g of sucrose (C12H22O11). Solution: (i) Molar mass of CS2 = 12 + 2(32) = 76 g/mol. Moles = 38/76 = 0.499 moles (about 0.5 mole). Molecules = 0.499 x 6.02 x 10^23 = 3.0 x 10^23 molecules. (ii) Molar mass of sucrose = 12(12) + 22(1) + 11(16) = 342 g/mol. Moles = 68.4/342 = 0.199 moles (about 0.2 mole). Molecules = 0.199 x 6.02 x 10^23 = 1.19 x 10^23 molecules.

Numerical 2: 1.6g of a sample of gas occupies a volume of 1.12 dm3 at STP. Determine the molar mass of the substance. Solution: At STP one mole of a gas occupies 22.4 dm3. Moles of gas = 1.12/22.4 = 0.05 mol. Molar mass = mass / moles = 1.6/0.05 = 32 g/mol.

Numerical 3: Calculate (i) number of moles in 6.4g of SO2 (ii) mass in grams of 4.5 moles of ethyne (C2H2) (iii) volume in cm3 of 38.4g O2 gas at STP (iv) number of molecules of 126g water (v) mass in grams of 4.8 x 10^24 atoms of sodium (vi) number of formula units in 333g of CaCl2. Solution: (i) Moles = 6.4/64 = 0.1 mol. (ii) Molar mass of C2H2 = 26 g/mol, mass = 4.5 x 26 = 117g. (iii) Moles of O2 = 38.4/32 = 1.2 mol; volume = 1.2 x 22400 = 26880 cm3 at STP. (iv) Moles of water = 126/18 = 7 mol; molecules = 7 x 6.02 x 10^23 = 42.14 x 10^23. (v) Moles of Na = 4.8 x 10^24 / 6.02 x 10^23 = 7.97 mol; mass = 7.97 x 23 = 183.39g. (vi) Moles of CaCl2 = 333/111 = 3 mol; formula units = 3 x 6.02 x 10^23 = 18.06 x 10^23.

Rounding off Data

LESSON 2

Experimental measurements often give numbers with more digits than are significant. Rounding of data is a method of reducing the figures in a given numerical value to the desired significant figures.

1. Rules of Rounding off Data

- Rounding means reducing a number to the desired significant figures and adjusting the last reported digit.
- If the digit to be dropped is less than 5, the last retained digit stays unchanged.

- If the digit to be dropped is greater than 5, the last retained digit is increased by one.
- If the digit to be dropped is exactly 5, the last retained digit is increased by one only if it is odd, and left unchanged if it is even.

Exponential Notation

LESSON 3

Very large and very small numbers in chemistry, such as Avogadro's number, are conveniently written in exponential notation.

1. Exponential Notation

Exponential notation is a shorthand mathematical form in which a number is written as a coefficient multiplied by 10 raised to some power. The decimal point is moved so that only one non-zero digit is left before it, and the power of 10 counts the places moved.

- 3652 = 3.652 x 10^3
- 0.0231 = 2.31 x 10^-2
- 0.000072 = 7.2 x 10^-5

2. Converting Back to Simple Numbers

A negative power moves the decimal point to the left and a positive power moves it to the right.

- 3.26 x 10^-3 = 0.00326
- 1.921 x 10^2 = 192.1
- 1.02 x 10^5 = 102000

Limiting Reactant and its Calculations

LESSON 4

In a reaction mixture the reactants are not always present in exactly the amounts required by the balanced equation. The reactant that runs out first controls how much product can form.

1. Limiting Reactant

The limiting reactant is the reactant which is consumed entirely when a chemical reaction is completed, while the other reactant is left partly unused (reactant in excess). It controls the amount of product because the reaction stops as soon as the limiting reactant is used up, even though the excess reactant is still present. Therefore the quantity of product is always calculated from the amount of the limiting reactant. For example, if hydrogen is limiting in the burning of hydrogen in oxygen, the water formed depends only on the hydrogen available.

2. Worked Numericals

Numerical 1: Ammonia gas can be produced by heating together solid NH4Cl and Ca(OH)2: 2NH4Cl + Ca(OH)2 -> 2NH3 + CaCl2 + 2H2O. If a mixture containing 100g of each of these solids is heated, determine the limiting reactant and the mass of NH3 gas produced. Solution: Moles of NH4Cl = 100/53.5 = 1.87 mol; moles of Ca(OH)2 = 100/74 = 1.35 mol. The equation needs 2 moles NH4Cl for 1 mole Ca(OH)2, so 1.87 mol NH4Cl needs only 0.935 mol Ca(OH)2, which is available in excess. Hence NH4Cl is the limiting reactant. NH3 formed = 1.87 mol, and mass of NH3 = 1.87 x 17 = 31.7g.

Numerical 2: Silver sulphide (Ag2S) is an anti-microbial agent. In an experiment 24.8g Ag2S is reacted with excess hydrochloric acid: Ag2S(s) + 2HCl(aq) -> 2AgCl(s) + H2S(g). Calculate (i) the mass of AgCl formed (ii) the volume of H2S produced at STP (atomic mass of Ag = 108 and S = 32). Solution: Molar mass of Ag2S = 2(108) + 32 = 248 g/mol, so moles = 24.8/248 = 0.1 mol. (i) AgCl formed = 2 x 0.1 = 0.2 mol; molar mass of AgCl = 108 + 35.5 = 143.5 g/mol, so mass = 0.2 x 143.5 = 28.7g. (ii) H2S formed = 0.1 mol, so volume at STP = 0.1 x 22.4 = 2.24 dm3.

Theoretical Yield, Practical Yield and Percent Yield

LESSON 5

The amount of product predicted by stoichiometry is rarely the amount actually collected in the laboratory. This lesson defines the three yield terms and how percent yield is calculated.

1. Theoretical Yield

Theoretical yield is the maximum amount of product calculated from a balanced chemical equation, assuming complete conversion of the limiting reactant.

2. Actual (Practical) Yield

Actual or practical yield is the amount of product actually obtained during the experiment. Practical yield is often less than theoretical yield because the reaction may not go to completion, side reactions may consume reactants, some product is lost during separation and purification, and the reaction may be reversible.

3. Percent Yield

Percent yield = (Practical yield / Theoretical yield) x 100

4. Worked Numerical

Numerical: Aluminum chloride is used in the manufacturing of rubber. It is produced by allowing aluminum to react with Cl2 gas at 650 degree C: 2Al(s) + 3Cl2(g) -> 2AlCl3(l). When 160g aluminum reacts with excess of chlorine, 650g of AlCl3 is produced. What is the percentage yield of AlCl3? Solution: Moles of Al = 160/27 = 5.93 mol, so theoretical moles of AlCl3 = 5.93 mol. Theoretical mass = 5.93 x 133.5 = about 791g. Percent yield = (actual yield / theoretical yield) x 100 = (650/791) x 100 = 82.17%.

Atomic Structure

UNIT 2

Subatomic Particles and their Characteristics

LESSON 6

More than a hundred subatomic particles have been discovered in an atom, such as electron, proton, neutron, positron, mesons and hyperons, however only electron, proton and neutron are considered to be the fundamental particles of an atom which play an important role in the physical and chemical properties of elements. Almost all of the mass of an atom exists in the nucleus, and except protium (isotope of hydrogen), nuclei of all other atoms contain neutrons.

1. Discovery Timeline

- Proton was discovered in 1886.
- Electron was discovered in 1897.
- Rutherford discovered the nucleus in 1911 and laid the foundation that the atom consists of an extremely charged region called nucleus wherein most of the mass of an atom is concentrated, while electrons revolve in the extra-nuclear part.
- James Chadwick discovered the neutron in 1932 and was awarded the Nobel Prize in 1935 for this discovery.
- Millikan, in 1909, performed the oil drop experiment and found that the charge on every oil drop was an integral multiple of 1.6 x 10^-19 Coulomb, which he deduced to be the charge of Thomson's electrons.

2. Electrons

The electron was discovered by J.J. Thomson in 1897 A.D. while studying the cathode rays experiment; he called the cathode rays beam as electrons. Thereafter, Millikan with his oil drop experiment calculated the charge on an electron. Electrons revolve in orbits and occupy about 100,000 times greater volume than the nucleus but form less than one percent total mass of an atom. They carry negative charge whose magnitude is equal to the positive charge of protons, and this attractive force (electrostatic force) keeps electrons constantly moving around the nucleus. The mass of an electron is nearly 1836 times less than a proton and 1839 times less than a neutron. Electronic configuration refers to the presence of electrons in various shells and sub-shells; this arrangement affects atomic stability, melting points, boiling points and density, and electrons are also involved in chemical bond formation.

3. Protons

Protons are positively charged subatomic particles which are located in the nuclei of all elements. They were discovered by Goldstein (1886 A.D.), a German physicist, by using a perforated cathode in Crook's discharge tube; Rutherford called these positively charged particles "protons". With respect to mass and charge, the protons of all elements are identical to each other. Atomic number of an element is related only to protons, while mass number is the sum of protons and neutrons. About 99.94% mass of an atom exists in the nucleus. Protons and electrons have the same magnitude of charge but have different masses, and all elements have different numbers of protons.

4. Neutrons

The discovery of the neutron was made by James Chadwick (1932 A.D.) in an artificial radioactivity experiment. He bombarded alpha particles from polonium on beryllium, a stable element, and noted that some highly penetrating neutral particles were produced; these particles were named as neutrons: Be-9 + He-4 -> C-12 + n-1 [neutron]. Like protons, neutrons are also part of the atomic mass, but the neutron has no charge. Neutrons of all elements have the same nature and mass, and they are bound with each other and with protons by nuclear forces.

Bohr's Atomic Theory

LESSON 7

Bohr's theory explains the structure of one-electron species such as H, He+ and Li+2 (it is not applicable to multi-electron species like H-, which has two electrons). Its ideas explain the line spectrum of hydrogen: each element gives a unique line spectrum, so the spectrum acts as a fingerprint for the identification of elements.

1. Postulates of Bohr's Atomic Theory

- Electrons revolve around the nucleus in fixed circular orbits of definite energy without radiating energy.
- The angular momentum of the electron is an integral multiple of h/2pi.
- Energy is absorbed when an electron jumps to a higher orbit and emitted when it falls to a lower orbit, with frequency given by E2 - E1 = hv.

2. Frequency of the Emitted Radiation

The energy of the electron in orbit n is En = -me^4/(8e0^2 h^2 n^2). For a jump from n2 to n1, the energy emitted is DE = E(n2) - E(n1) = (me^4/8e0^2 h^2)(1/n1^2 - 1/n2^2). Since DE = hv, the frequency is v = (me^4/8e0^2 h^3)(1/n1^2 - 1/n2^2), which explains the lines of the hydrogen spectrum. The radius of an orbit is given by r = 0.529 x n^2/Z in Angstrom; the radius of the first orbit of hydrogen is 0.529 A.

3. Hydrogen Spectrum in Terms of Bohr's Theory

In the ground state the electron of hydrogen stays in the first orbit. When energy is supplied, it jumps to higher orbits (n = 2, 3, 4 ...). When it falls back, it emits radiation of fixed frequency, producing a line spectrum. Falls to n = 1 give the Lyman series in the ultraviolet region, falls to n = 2 give the Balmer series in the visible region, and falls to n = 3, 4 and 5 give the Paschen, Brackett and Pfund series in the infrared region. Each line corresponds to a definite energy difference between two orbits.

4. Defects of Bohr's Atomic Model

- It does not explain the spectra of atoms having more than one electron, such as helium.
- It cannot explain the fine structure of spectral lines observed under high resolution, nor the splitting of lines in electric and magnetic fields (Stark and Zeeman effects).
- It treats the electron only as a particle and ignores its wave nature.
- It violates Heisenberg's uncertainty principle by fixing both the position and momentum of the electron.
- It gives no satisfactory explanation of chemical bonding and molecular formation.

5. Worked Numericals

Numerical 1: Calculate the radius of hydrogen in the 3rd orbit (Bohr constant for hydrogen is 0.529 A). Solution: rn = 0.529 x n^2/Z. For hydrogen Z = 1 and n = 3, so r3 = 0.529 x 9/1 = 4.761 A. Hence the radius of the third orbit of hydrogen is about 4.76 A.

Numerical 2: A photon of wave number 23 x 10^5 m^-1 is emitted when an electron undergoes a transition from a higher energy orbit to n = 2. Determine the orbit from which the electron falls and the spectral series (Rydberg constant = 1.09678 x 10^7 m^-1). Solution: Wave number = RH (1/n1^2 - 1/n2^2) with n1 = 2. So 23 x 10^5 = 1.09678 x 10^7 (1/4 - 1/n2^2), giving (1/4 - 1/n2^2) = 0.2097. Then 1/n2^2 = 0.25 - 0.2097 = 0.0403, so n2^2 = 24.8 and n2 = about 5. The electron falls from the 5th orbit to the 2nd orbit; since the lower orbit is n = 2, the line belongs to the Balmer series and appears in the visible region.

Planck's Quantum Theory

LESSON 8

Planck's quantum theory replaced the idea of continuous energy flow with the idea that energy is emitted or absorbed in small packets. Bohr used this idea when he related the energy difference between orbits to the frequency of radiation: E2 - E1 = hν.

1. Quanta and Photons

- Energy is not emitted continuously but in discrete packets called quanta; a quantum of light is called a photon.
- The energy of one quantum is directly proportional to the frequency of the radiation: E = hν, where h is Planck's constant.
- This is why each spectral line has a definite frequency: it corresponds to one definite energy difference DE = hν between two orbits.

2. Continuous and Line Spectrum

Continuous Spectrum	Line Spectrum
Produced when white light passes through a prism	Consists of separate bright or dark lines with empty spaces between them
A band of seven colours (VIBGYOR) which diffuse into each other with no distinct boundary line	Each line has a definite wavelength
Given by hot solids and liquids	Given by gases and vapours in a discharge tube
Not characteristic of an element	Characteristic of elements and used for their identification

X-Rays

LESSON 9

X-rays are short wavelength, highly penetrating electromagnetic radiations.

1. Production of X-Rays

They are produced when fast-moving electrons (cathode rays) strike a metal target of high atomic mass, such as tungsten, in a discharge tube; part of their kinetic energy changes into X-rays.

2. Properties of X-Rays

- They travel in straight lines at the speed of light.
- They are not deflected by electric or magnetic fields.
- They penetrate flesh but are stopped by bones and heavy metals.
- They affect photographic plates.

3. Uses of X-Rays

- In medical diagnosis and surgery.
- In industry for detecting defects in materials.
- In the study of crystal structure.
- In the treatment of some diseases.

Radioactivity

LESSON 10

Radioactive substances emit three kinds of rays — alpha, beta and gamma — which differ in charge, penetrating power and ionizing power.

1. Properties of Alpha, Beta and Gamma Rays

Alpha Rays	Beta Rays	Gamma Rays
Consist of positively charged particles (helium nuclei)	Consist of negatively charged particles (electrons)	Electromagnetic radiations of very short wavelength with no charge
Deflect towards the negative pole in an electric field	Deflect towards the positive pole in an electric field	Pass through electric and magnetic fields without deflection
Low penetrating power	Greater penetrating power than alpha rays	Highest penetrating power
High ionizing power	Moderate ionizing power	Low ionizing power

Quantum Numbers and Orbitals

LESSON 11

In the quantum mechanical picture an electron is not in a fixed orbit but in an orbital, and its state is completely described by four quantum numbers.

1. The Four Quantum Numbers

- **Principal quantum number (n):** describes the size and energy of the shell and takes values 1, 2, 3 ....
- **Azimuthal quantum number (l):** describes the shape of the orbital and ranges from 0 to (n - 1). For a 3s orbital, n = 3 and l = 0.
- **Magnetic quantum number (m):** describes the orientation of the orbital in space and ranges from -l to +l.

- **Spin quantum number (s):** describes the spin of the electron and is +1/2 or -1/2.

For n = 2: l = 0, 1; for l = 0, m = 0; for l = 1, m = -1, 0, +1; and for each electron s = +1/2 or -1/2.

2. Shapes of Orbitals

- For l = 0 the orbital is an s orbital, which is spherical in shape, with the electron density distributed equally in all directions around the nucleus.
- For l = 1 the orbital is a p orbital, which is dumbbell shaped, having two lobes on opposite sides of the nucleus separated by a nodal plane. There are three p orbitals (px, py, pz) oriented along the three axes.
- The size of both types of orbital increases with the principal quantum number n.

3. Orbit versus Orbital

Orbit	Orbital
A well-defined circular path around the nucleus in which an electron revolves, as proposed by Bohr	A three-dimensional region of space around the nucleus where the probability of finding an electron is maximum (about 95%)
Position and momentum of the electron can be known at the same time	Based on quantum mechanics and the uncertainty principle
Two-dimensional concept with a fixed radius	Has a definite shape (spherical, dumbbell, etc.) but no fixed path

Electronic Configuration

LESSON 12

Electronic configuration is the arrangement of electrons in the shells and sub-shells of an atom. Three rules govern this arrangement.

1. Pauli's Exclusion Principle

In an atom no two electrons can have the same set of four quantum numbers; at least one quantum number must be different. Hence an orbital can hold at most two electrons, and they must have opposite spins, illustrated as one up arrow and one down arrow in the same box.

2. Hund's Rule of Maximum Multiplicity

When degenerate orbitals are available, electrons fill them singly with the same spin direction so as to give the maximum number of unpaired electrons. For example, nitrogen has the configuration 1s2 2s2 2p1x 2p1y 2p1z and not 1s2 2s2 2p2x 2p1y, because the three p electrons occupy separate p orbitals. This is determined by Hund's rule.

3. Aufbau Principle and the (n + l) Rule

Electrons fill orbitals in order of increasing energy, decided by the (n + l) rule: the orbital with the lower (n + l) value is filled first. This explains why the filling of the 4s orbital takes place prior to 3d: for 4s, (n + l) = 4 + 0 = 4, while for 3d, (n + l) = 3 + 2 = 5, so 4s has lower energy and is filled first. This is why potassium and calcium place their electrons in 4s before 3d begins to fill in the transition metals. After 4d is filled, the next electron enters 5p.

4. Writing Electronic Configurations

- Fe (Z = 26): 1s2 2s2 2p6 3s2 3p6 4s2 3d6
- Br- (Z = 35) has 35 + 1 = 36 electrons: 1s2 2s2 2p6 3s2 3p6 4s2 3d10 4p6, the same as krypton.
- Ca+2 (Z = 20) has 20 - 2 = 18 electrons: 1s2 2s2 2p6 3s2 3p6, the same as argon.

Isoelectronic species have the same number of electrons: Na+ and Ne, Na+ and F-, and Na+ and Mg+2 are isoelectronic pairs (each has 10 electrons), while Na (11 electrons) and Ca (20 electrons) are not.

Theories of Covalent Bonding and Shapes of Molecules

UNIT 3

Theories of Covalent Bond

LESSON 13

Two theories explain the covalent bond: valence bond theory (VBT), in which the bond is localized between two atoms, and molecular orbital theory (MOT), in which electrons occupy orbitals spread over the whole molecule.

1. Valence Bond Theory (VBT)

- A covalent bond is formed by the overlapping of half-filled atomic orbitals of two atoms, and the shared electron pair is localized between the two nuclei.
- The overlapping takes place in the direction of maximum electron density, and the atoms keep their identity in the molecule.
- Greater overlapping gives a stronger bond.

2. Sigma and Pi Bonds

End-to-end (head-on) overlapping along the internuclear axis, such as s-s, s-p and p-p axial overlap, produces a **sigma bond**, which is strong and allows free rotation. Side-by-side overlapping of parallel p orbitals above and below the axis produces a **pi bond**, which is weaker and restricts rotation about the bond. A sigma bond is stronger than a pi bond because the overlapping in a sigma bond is head-on and extensive, while in a pi bond the side-by-side overlapping is partial and less effective. In C2H4 there are 5 sigma bonds and 1 pi bond: the four C-H single bonds and one part of the C=C double bond are sigma, and the remaining part of the double bond is pi.

3. Molecular Orbital Theory (MOT)

In MOT, atomic orbitals combine to form molecular orbitals spread over the whole molecule. A bonding molecular orbital has lower energy than the atomic orbitals and holds the atoms together, while an antibonding molecular orbital has higher energy and tends to separate the atoms.

VBT	MOT
Bond formed by overlapping of atomic orbitals	Atomic orbitals combine to form molecular orbitals
Shared pair stays localized between the two atoms	Electrons occupy orbitals spread over the whole molecule

4. Molecular Orbital Diagram of O2

The molecular orbital diagram of O2 is obtained by combining the atomic orbitals of two oxygen atoms; the 16 electrons fill the molecular orbitals in order of increasing energy.

- **MO configuration:** (sigma1s)2 (sigma*1s)2 (sigma2s)2 (sigma*2s)2 (sigma2pz)2 (pi2px)2 (pi2py)2 (pi*2px)1 (pi*2py)1
- **Bond order** = (bonding electrons - antibonding electrons)/2 = (10 - 6)/2 = 2, so O2 has a double bond. (In N2, bonding electrons are 10 and antibonding 4, so bond order = (10 - 4)/2 = 3, a triple bond.)
- **Paramagnetic behaviour:** the two antibonding pi* orbitals each contain one unpaired electron. Because of these two unpaired electrons, O2 is paramagnetic, and this behaviour can only be explained by molecular orbital theory, not by valence bond theory.

Shape of the Molecules

LESSON 14

The shape of a molecule is decided by the repulsion between the electron pairs around its central atom, and by the hybridization of that atom.

1. VSEPR Theory

- The shape of a molecule is determined by the repulsion between electron pairs around the central atom.
- The pairs arrange themselves as far apart as possible to minimize repulsion.
- Repulsion decreases in the order: lone pair-lone pair > lone pair-bond pair > bond pair-bond pair.

2. Shapes Predicted by VSEPR Theory

Molecule	Bond Pairs	Lone Pairs	Shape
AlCl3	3	0	Planar trigonal
CCl4	4	0	Tetrahedral
PH3	3	1	Trigonal pyramidal
H2S	2	2	Angular (bent)
NH3	3	1	Trigonal pyramidal (bond angle 107 degree)
H2O	2	2	Bent (bond angle 104.5 degree)
CS2	2	0	Linear (bond angle 180 degree, the maximum)
BF3	3	0	Planar trigonal (bond angle 120 degree)

3. Hybridization

Hybridization is the mixing of atomic orbitals of nearly equal energy to produce an equal number of new orbitals of the same energy, called hybrid orbitals.

- **sp3 hybridization in CH4:** the 2s and three 2p orbitals of carbon mix to form four sp3 hybrid orbitals directed towards the corners of a tetrahedron; each overlaps with the 1s orbital of hydrogen, giving a tetrahedral molecule with 109.5 degree angles.
- **sp2 hybridization in C2H4:** each carbon mixes its 2s and two 2p orbitals to form three sp2 hybrid orbitals in one plane at 120 degree, used for sigma bonds with two hydrogens and the other carbon. The unhybridized p orbital on each carbon overlaps sideways to form the pi bond of the C=C double bond.
- **Expected hybridization:** BeCl2 — sp, linear; BF3 — sp2, planar trigonal; CH4 — sp3, tetrahedral; C2H2 — each carbon is sp, linear.

Bond Characteristics

LESSON 15

Bond energy, bond length and bond order describe the strength and size of a covalent bond, and together they decide many properties of the molecule.

1. Bond Energy

Bond energy is the amount of energy required to break one mole of a particular type of bond in a molecule; it is measured in KJ/mol and shows the strength of the bond. The H-F bond is very short and strongly polar due to the small size and high electronegativity of fluorine, so it has the highest bond energy among bonds like Cl-Cl, H-O and H-N. The bond energy of molecules possessing multiple bonds is high, because a multiple bond (double or triple) contains one sigma and one or two pi bonds, more energy is needed to break all of them together, and the atoms are also held closer.

2. Bond Length

Bond length is the distance between the nuclei of two bonded atoms in a molecule, measured in Angstrom (A). It depends on the size of the atoms and the bond order: a higher bond order means a shorter bond length. For example, a C-C single bond is longer than a C=C double bond.

3. Bond Order

Bond order = (bonding electrons - antibonding electrons)/2. N2 has bond order 3 (triple bond) and O2 has bond order 2 (double bond).

4. Ionic Character

HF has greater ionic character than HCl because fluorine is much more electronegative than chlorine, so the electronegativity difference between H and F is larger than between H and Cl; hence the H-F bond is more polar.

Effect of Bonding on Physical and Chemical Properties

LESSON 16

The type of bonding and the shape of a molecule decide its polarity, and polarity in turn decides properties such as solubility.

1. Dipole Moment and Geometry

The dipole moment of water is 1.85D but CO2 has zero dipole moment. Water has a bent shape, so the dipoles of the two O-H bonds do not cancel; they add up to give a resultant dipole moment of 1.85D, making water polar. Carbon dioxide is a linear molecule (O=C=O), in which the two equal C=O bond dipoles point in exactly opposite directions, so they cancel each other completely and the net dipole moment of CO2 is zero. CCl4 also has zero dipole moment: it is a regular tetrahedron and the four equal C-Cl bond dipoles cancel each other. This shows that dipole moment depends not only on bond polarity but also on molecular geometry. The Debye is the practical unit of dipole moment: 1D = 3.33 x 10^-30 Cm.

2. Like Dissolves Like

Oil is insoluble in water but soluble in hexane. Oil is a non-polar covalent substance, while water is a polar solvent. According to the rule "like dissolves like", the polar water molecules attract each other strongly through hydrogen bonding and do not allow the non-polar oil molecules to mix, so oil forms a separate layer. Hexane is also non-polar, and the weak London forces between oil and hexane molecules are comparable with the forces within each liquid, so oil dissolves in hexane, just as non-polar substances dissolve in non-polar solvents.

State of Matter I: Gas

UNIT 4

Kinetic Molecular Theory of Gases

LESSON 17

The behaviour of gases is explained by the kinetic molecular theory, which pictures a gas as a large number of tiny particles in rapid motion.

1. Main Postulates of Kinetic Molecular Theory

- A gas consists of a large number of tiny particles (molecules) which are far apart, so the actual volume of the molecules is negligible compared with the volume of the gas.
- The molecules are in continuous rapid motion in all directions and collide with each other and with the walls of the container; these collisions are perfectly elastic.
- There are no forces of attraction between the molecules.
- The average kinetic energy of the molecules is directly proportional to the absolute temperature.
- The pressure of the gas is produced by the collisions of molecules with the walls of the container.

2. Pressure and Its Units

Pressure is the force applied per unit area on the surface of an object; for a gas it results from collisions of molecules with the walls of the container. Its SI unit is the pascal (N/m2), and it is also measured in atmosphere, torr and mm of Hg, where 1 atm = 760 torr = 760 mmHg = 101325 pascal. Molar mass is a fixed property of a gas and does not depend on temperature.

Absolute Temperature and Charles's Law

LESSON 18

Charles's law connects the volume of a gas with its temperature and leads to the idea of absolute zero.

1. Charles's Law

Charles's law states that the volume of a given mass of a gas is directly proportional to its absolute temperature when pressure is kept constant, V proportional to T. It is used to define absolute zero, the temperature at which the volume of a gas would become zero.

2. Absolute Zero

Absolute zero is the lowest possible temperature, 0 K or -273.15 degree C, at which the volume of an ideal gas would become zero. In practice all gases liquefy before reaching it, and at absolute zero the kinetic energy and motion of the particles are at their minimum. By Charles's law the volume of a gas becomes theoretically zero at 0 K, not at 0 degree C or 273 K.

3. Effect of Temperature and Pressure Together

From V proportional to T/P: if the Kelvin temperature of an ideal gas is doubled and the pressure is reduced to one half, doubling T doubles V and halving P doubles it again, so the volume becomes four times.

Avogadro's Law

LESSON 19

Avogadro's law relates the volume of a gas to the number of its particles.

1. Statement of Avogadro's Law

Avogadro's law states that equal volumes of all gases contain an equal number of moles (or molecules) at the same temperature and pressure; the volume of a gas is directly proportional to its number of moles when temperature and pressure are constant, V proportional to n.

2. Molar Volume

It follows that one mole of every ideal gas occupies the same volume at STP: the molar volume of oxygen gas is 22.4 dm3 at 0 degree C and 1 atm. Molar volume increases with temperature and decreases with pressure, so it is maximum at the highest temperature and lowest pressure, e.g. at 25 degree C and 1 atm rather than at 0 degree C and 2 atm.

Ideal Gas Equation

LESSON 20

Boyle's law, Charles's law and Avogadro's law combine into a single equation of state for an ideal gas.

1. Derivation of the General Gas Equation

Boyle's law gives V proportional to 1/P at constant T, and Charles's law gives V proportional to T at constant P. Combining them, V proportional to T/P, or PV proportional to T. For n moles this becomes:

PV = nRT

where R is the universal gas constant.

2. Value of R

At STP one mole occupies 22.4 dm3 at 1 atm and 273K, so R = PV/nT = (1 x 22.4)/(1 x 273) = 0.0821 atm dm3/mol.K. In SI units, R = (101325 Pa x 0.0224 m3)/(1 x 273) = 8.314 J/mol.K.

3. Worked Numericals

Numerical 1: Gorakh hill station is the coldest area of Sindh province and has 83 KPa barometer pressure. What will be the pressure of this area in psi and atm units? Solution: 1 atm = 101.325 kPa, so P = 83/101.325 = 0.819 atm. In psi: 1 atm = 14.7 psi, so P = 0.819 x 14.7 = about 12.04 psi.

Numerical 2: Calculate the volume occupied by 8g of methane gas at 40 degree C and 842 torr pressure. Solution: Moles of CH4 = 8/16 = 0.5 mol. Pressure = 842/760 = 1.108 atm, and T = 40 + 273 = 313K. V = nRT/P = (0.5 x 0.0821 x 313)/1.108 = 11.58 dm3.

Numerical 3: At 35 degree C, oxygen gas in a cylinder has 456 cm3 volume and 0.85 atm pressure. Calculate the pressure when this oxygen gas is transferred to a 10 dm3 cylinder and cooled to 20 degree C. Solution: V1 = 0.456 dm3, T1 = 308K, T2 = 293K, V2 = 10 dm3. Using the combined gas law, P2 = P1V1T2/(V2T1) = (0.85 x 0.456 x 293)/(10 x 308) = 0.037 atm. The pressure falls greatly because the gas is expanded into a much larger volume.

Numerical 4: Four containers of equal volume are filled as follows: (i) 2.0g of H2 at 0 degree C (ii) 1.0g of H2 at 273 degree C (iii) 24g of O2 at 0 degree C (iv) 16g of CH4 at 273 degree C. Which container is at the greatest pressure and which at the lowest? Solution: For equal volumes, pressure is proportional to n x T. (i) n = 1 mol, T = 273K, nT = 273. (ii) n = 0.5 mol, T = 546K, nT = 273. (iii) n = 24/32 = 0.75 mol, T = 273K, nT = 204.75. (iv) n = 1 mol, T = 546K, nT = 546. Container (iv) has the greatest nT, so it is at the greatest pressure; container (iii) has the smallest nT, so it is at the lowest pressure.

Deviation from Ideal Gas Behaviour

LESSON 21

No real gas obeys PV = nRT under all conditions. Real gases approach ideal behaviour only when their molecules are far apart and intermolecular forces are negligible.

1. Ideal Gas versus Real Gas

Ideal Gas	Real Gas
Obeys the gas laws and PV = nRT at all temperatures and pressures	Obeys the van der Waals equation; shows ideal behaviour only at high temperature and low pressure
Molecules assumed to have no volume	Molecules occupy a finite volume
No intermolecular attractions assumed	Attractive forces exist between molecules

2. Causes of Deviation

The causes of deviation are the two faulty assumptions of the kinetic theory: real gas molecules do occupy a finite volume, and attractive forces do exist between them.

- **At low temperature:** the molecules move slowly, so attractions pull them together and the volume and pressure are less than expected.
- **At high pressure:** the molecules are crowded close together, so their own volume cannot be neglected and the gas is less compressible than an ideal gas.
- **Ideal behaviour:** a real gas reaches ideal behaviour at high temperature and low pressure. Among all real gases, helium acts most like an ideal gas at room temperature.

Van der Waals Equation

LESSON 22

In order to modify the two conflicting postulates of the kinetic molecular theory regarding volumes of gas molecules and intermolecular attraction of gas molecules, the Dutch scientist J.D. van der Waal (1873) gave a mathematical solution by the correction in molecular volume and intermolecular forces. This is known as the van der Waal's equation.

1. Volume Correction

When pressure is applied to a gas, the molecules come closer to each other, and by continuous increase in pressure a point is reached when molecules cannot be compressed further, since repulsive forces are created. This indicates that gas molecules have a definite volume. Keeping in view the definite volume of gas molecules, van der Waal calculated the actual (free) volume of a gas as:

V = V(vessel) - nb

where V = free volume, V(vessel) = volume of the vessel in which gas molecules are present, n = number of moles, and b = excluded volume of gas molecules per mole. The value of b depends upon the size of gas molecules; b is not the actual volume of the gas but is assumed to be four times the actual volume of the molecules: b = 4V(molecule).

2. Pressure Correction

A molecule in the interior of the gas is completely surrounded and attracted by the other gas molecules, so the resultant attractive force on it is zero. However, as a molecule approaches to strike the wall of the vessel, it experiences a net inward pull due to the attractive forces of the molecules behind it, so the pressure produced on the wall is a little less than the pressure of an ideal gas molecule. The pressure drop PL is proportional to the number of particles per unit volume: PL = an^2/V^2, where a is a constant. Hence the ideal pressure is Pi = P + an^2/V^2.

3. The van der Waals Equation

Putting the corrected pressure and volume in the general gas equation:

(P + an^2/V^2)(V - nb) = nRT

If n = 1, the equation becomes (P + a/V^2)(V - b) = RT. Here a and b are van der Waal's constants. Since P = an^2/V^2 gives a = PV^2/n^2, the unit of a is atm dm6 mole-2; b represents the volume per mole of gas, so its unit is dm3/mol.

4. van der Waal's Constants for Some Common Real Gases

Gas	a (atm dm6 mole-2)	b (dm3 mole-1)
Hydrogen	0.0247	0.0266
Nitrogen	1.390	0.0391
Oxygen	1.360	0.0318
Methane	2.253	0.0428
Carbon dioxide	3.590	0.0428
Ammonia	4.170	0.0371
Sulphur dioxide	6.170	0.0564
Chlorine	6.493	0.0562

Dalton's Law of Partial Pressure

LESSON 23

When two or more gases are mixed in one vessel, each gas contributes its own share to the total pressure.

1. Statement of Dalton's Law

Dalton's law states that the total pressure of a mixture of two or more non-reacting gases in a fixed volume enclosed container is equal to the sum of the partial pressures of the individual gases:

P(total) = P1 + P2 + P3 + ...

The partial pressure of a gas is the pressure it would exert if it occupied the vessel alone, and it can be calculated from its mole fraction multiplied by the total pressure.

2. Practical Applications of Dalton's Law

- Collecting a gas over water and correcting its pressure for water vapour: the total pressure is the sum of the partial pressure of the gas and the vapour pressure of water.
- Calculating pressures in gas mixtures used in deep-sea diving.
- Working out the composition of air and respiratory gases.
- In industry, wherever gases are stored and mixed in cylinders.

3. Worked Numericals

Numerical 1: A 500 cm3 vessel contains H2 gas at 400 torr pressure and another 1 dm3 vessel contains O2 gas at 600 torr pressure. If under similar conditions of temperature these gases are transferred to a 2 dm3 empty vessel, calculate the pressure of the mixture. Solution: On transfer, each gas expands into 2 dm3, so find its partial pressure by Boyle's law. For H2: P = 400 x 0.5/2 = 100 torr. For O2: P = 600 x 1/2 = 300 torr. By Dalton's law, total pressure = 100 + 300 = 400 torr.

Numerical 2: 40 dm3 of hydrogen gas was collected over water at 831 torr pressure at 23 degree C. What would be the volume of dry hydrogen gas at standard conditions? The vapour pressure of water at 23 degree C is 21 torr. Solution: Pressure of dry H2 = 831 - 21 = 810 torr. T1 = 296K; at standard conditions P2 = 760 torr and T2 = 273K. V2 = P1V1T2/(T1P2) = (810 x 40 x 273)/(296 x 760) = about 39.3 dm3.

Graham's Law of Diffusion and Effusion

LESSON 24

Gases spread out and escape through small openings at rates that depend on their molar masses.

1. Diffusion and Effusion

- **Diffusion** is the spontaneous mixing of gases (or particles) due to their random molecular motion, so that they spread uniformly throughout the available space.
- **Effusion** is the escape of a gas through a tiny hole or porous material into a region of lower pressure.

2. Graham's Law

Graham's law states that the rate of diffusion (or effusion) of a gas is inversely proportional to the square root of its density or molar mass at constant temperature and pressure. Thus for two gases:

r1/r2 = sqrt(M2/M1)

Lighter gases like hydrogen diffuse much faster than heavier gases like oxygen: sqrt(32/2) = 4, so H2 diffuses 4 times faster than O2, a ratio of 4:1. C3H8 and CO2 diffuse at the same rate because both have molar mass 44 g/mol.

3. Worked Numericals

Numerical 1: Compare the rates of diffusion of the following pairs of gases: (1) H2 and D2 (2) He and SO2 (3) SF6 and SO2. Solution: By Graham's law, rate ratio = sqrt(M of heavier gas / M of lighter gas). (1) H2 (M = 2) and D2 (M = 4): r(H2)/r(D2) = sqrt(4/2) = 1.41. (2) He (M = 4) and SO2 (M = 64): r(He)/r(SO2) = sqrt(64/4) = 4. (3) SF6 (M = 146) and SO2 (M = 64): r(SO2)/r(SF6) = sqrt(146/64) = 1.51, i.e. the lighter SO2 diffuses 1.51 times faster.

Numerical 2: If 16 cm3 of hydrogen effuses in 30 sec from a porous material, what volume of SO2 will effuse in the same time under similar conditions? Solution: Volume effused in the same time is inversely proportional to the square root of molar mass. V(SO2) = V(H2) x sqrt(M of H2 / M of SO2) = 16 x sqrt(2/64) = 16/5.66 = 2.83 cm3. Only about 2.83 cm3 of SO2 effuses in 30 seconds because SO2 is much heavier than hydrogen.

Liquefaction of Gases

LESSON 25

Gases can be turned into liquids by compressing them and cooling them to very low temperatures; air itself can be liquefied in this way.

1. Liquid Air

Liquid air is ordinary air which, on compression and cooling to an extremely low temperature (about -190 degree C), becomes liquefied. It is a pale blue liquid mixture mainly of liquid nitrogen and liquid oxygen. Liquid air must be stored in special insulated vessels because it evaporates very quickly.

2. Uses of Liquid Air

- It provides a very low temperature for scientific research and refrigeration.
- Fractional distillation of liquid air gives nitrogen, oxygen and noble gases for industry and hospitals.
- Liquid oxygen obtained from it is used in rockets, welding and medical treatment.

Fourth State of Matter: Plasma

LESSON 26

Beyond solid, liquid and gas there is a fourth state of matter called plasma, found in the sun and in many modern devices.

1. What is Plasma?

Plasma is the fourth state of matter, consisting of a mixture of ions, electrons and un-ionized atoms and molecules. It is formed when a gas is heated to a very high temperature or subjected to a strong electric field, so that its atoms become ionized.

2. Significance of Plasma in Daily Life

- Fluorescent tubes and neon sign lights work on plasma.
- Plasma television screens use it to produce pictures.
- Plasma torches are used for cutting metals.
- The sun and other stars consist of plasma.
- It is being studied for producing electricity by nuclear fusion.

State of Matter II: Liquids

UNIT 5

Kinetic Molecular Interpretation of Liquids

LESSON 27

In liquids the molecules are close together but not fixed; they are in constant motion and can slide over one another. This picture explains the familiar properties of liquids.

1. Kinetic Interpretation of Liquid Properties

- **Diffusion:** liquid molecules are in constant motion and slowly intermingle with the molecules of another liquid, though diffusion is slower than in gases because the molecules are closer together.
- **Compression:** liquid molecules are already closely packed with very little empty space, so liquids are almost incompressible.
- **Expansion:** on heating, the kinetic energy of the molecules increases and they move slightly apart, so liquids expand much less than gases.
- **Molecular motion:** liquid molecules do not have fixed positions; they slide over one another, which is why liquids can flow and take the shape of the container, while they possess a definite volume.

Inter-Molecular Forces in Liquids

LESSON 28

The properties of a liquid are governed by the forces of attraction between its molecules. There are three major kinds of intermolecular forces in liquids.

1. Kinds of Intermolecular Forces

- **Dipole-dipole interactions:** attractions between the positive end of one polar molecule and the negative end of another.
- **London dispersion forces:** weak forces due to temporary dipoles, present in all molecules; the smallest molecule F2 has the weakest London forces, and a non-polar molecule experiences only London forces, which increase with molecular size.
- **Hydrogen bonding:** the strongest of the three, found in molecules like H2O, NH3 and HF where hydrogen is bonded to F, O or N (it is absent in CH4, because carbon is not a highly electronegative atom).

2. Intermolecular Forces in HCl

HCl is a polar molecule with a permanent dipole, positive on hydrogen and negative on chlorine. In liquid HCl the positive end of one molecule attracts the negative end of a neighbouring molecule, giving dipole-dipole interactions. In addition, weak London dispersion forces act between all the molecules due to temporary dipoles. Hydrogen bonding of the strong type (as in water) is absent in HCl because chlorine is large and less electronegative in effect.

3. Hydrogen Bond

A hydrogen bond is the force of attraction between a partially positively charged hydrogen atom in one molecule and a lone pair of a highly electronegative atom (F, O or N) in another molecule. It is established when hydrogen is covalently bonded to such an electronegative atom, which pulls the shared electrons towards itself and leaves the hydrogen with a partial positive charge that attracts a neighbouring lone pair. In industry, hydrogen bonding explains the high solubility of substances like sugar and alcohol in water and is important in making adhesives, paints and fibres such as nylon. In biochemical processes it holds together the two strands of DNA, gives proteins their shapes, and is essential in enzyme action and the working of medicines.

4. Hydrogen Bonding in Water and the Anomalous Behaviour of Water

In water, the hydrogen atom of one molecule, being partially positive, is attracted to a lone pair of the oxygen atom of another molecule, forming a hydrogen bond. Each water molecule can form up to four hydrogen bonds, building an open, cage-like structure in ice. Because of this open structure, ice is less dense than liquid water and floats on it. Water also expands below 4 degree C, so its density is maximum at 4 degree C; this anomalous behaviour protects aquatic life, as ice forms only at the surface of ponds in winter. Hydrogen bonding is also why the boiling point of water (100 degree C) is much higher than that of H2S, which has no hydrogen bonding: the strong hydrogen bonds need extra heat to break.

5. Volatility and Intermolecular Forces

Volatility increases as intermolecular forces become weaker, because molecules escape more easily from the surface. Glycerin has three -OH groups per molecule and very extensive hydrogen bonding, so it is much less volatile than water. Petrol consists of non-polar hydrocarbon molecules held only by weak London forces, so its molecules escape even more easily than water molecules. Hence the order of volatility is petrol > water > glycerin, exactly following the decreasing strength of intermolecular forces.

Physical Properties of Liquids

LESSON 29

Surface tension, viscosity, evaporation and boiling are the observable physical properties of liquids, and all of them arise from intermolecular forces.

1. Surface Tension

Surface tension is the force responsible for the tension at the surface of a liquid; it is the energy required to stretch (increase) the surface by unit area. The surface molecules are pulled inwards by the molecules below them, so the surface behaves like a stretched elastic membrane; for example, a needle can float on water and insects can walk on it. Its unit is N/m (or J/m2). Surface tension decreases with rise in temperature and decreases when detergents or impurities are added, while strong intermolecular forces, such as hydrogen bonding and metallic bonding, give high surface tension as in water and mercury. Mercury shows the maximum surface tension because of its very strong cohesive (metallic) forces. A falling drop of a liquid is spherical because surface tension pulls the surface into the smallest possible area.

2. Meniscus

When cohesive forces within a liquid are stronger than adhesive forces between the liquid and glass, the liquid forms a convex (upward) meniscus, as mercury does. In mercury cohesive forces are stronger than adhesive forces with glass, so the surface curves upward. When adhesive forces are stronger, the meniscus is concave, as with water.

3. Viscosity: Why Honey is More Viscous than Water

Viscosity is the internal resistance to the flow of a liquid, caused by friction between its layers. Honey is more viscous than water because its molecules (large sugar molecules such as glucose and fructose) are big, irregular and strongly attracted to each other by extensive hydrogen bonding. These large molecules entangle and cannot slide past one another easily, so honey flows very slowly. Water molecules are small, and although they also hydrogen bond, the layers slip over each other with much less resistance. Viscosity also decreases on heating, which is why warm honey flows faster than cold honey.

4. Evaporation and Boiling Point

- Water spilled on the floor evaporates faster than the same amount of water in a container because spilled water has a much larger surface area, and evaporation takes place from the surface.
- Evaporation is a cooling process: the molecules with the highest kinetic energy escape, lowering the average energy and temperature of the remaining liquid. The cooling felt after a bath is due to evaporation of water from the skin.
- The boiling point of a liquid remains constant although heat is continuously supplied, because at the boiling point the supplied heat is used to overcome intermolecular forces (latent heat) instead of raising the temperature.
- Boiling points of different liquids may be different at the same external pressure due to intermolecular forces; stronger forces give a higher boiling point.
- In a pressure cooker the increased pressure raises the boiling point of water above 100 degree C, so cooking time is reduced.

Energetics of Phase Change

LESSON 30

Every change of state involves a definite amount of heat per mole of the substance.

1. Molar Heats of Phase Change

- **Molar heat of fusion** is the amount of heat required to convert one mole of a solid completely into liquid at its melting point; for water it is +6.02 KJ/mol.
- **Molar heat of vapourization** is the heat required to convert one mole of a liquid into its vapours at its boiling point; for water it is 40.7 KJ/mol.
- **Molar heat of sublimation** is the heat absorbed when one mole of a solid changes directly into vapours without passing through the liquid state. It is equal to the sum of the molar heat of fusion and the molar heat of vapourization.

Sublimation is an endothermic process, and evaporation also absorbs heat from the surroundings, so it is endothermic too (not exothermic); evaporation is the reverse process of condensation, occurs at all temperatures, and causes a cooling effect.

Liquid Crystals

LESSON 31

Liquid crystals are a state of matter between liquids and crystalline solids, widely used in display devices.

1. What are Liquid Crystals?

In liquid crystals the molecules keep an ordered arrangement like a crystal but can still flow like a liquid.

Compared with Liquids	Compared with Crystalline Solids
Their molecules are arranged in a regular direction, which makes them show optical properties such as double refraction	They have no rigid three-dimensional lattice
They flow and change shape like liquids	They melt over a range instead of a sharp melting point, and can flow and change shape

Liquid crystals are widely used in LCD displays of watches, calculators, ATM machines and computer screens.

State of Matter III: Solids

UNIT 6

Kinetic Molecular Interpretation of Solids

LESSON 32

In solids the particles are tightly packed in fixed positions with very strong attractive forces between them, which explains their rigidity and definite shape.

1. Why the Compressibility of Solids is Nearly Zero

In solids the particles are already tightly packed together with very small empty spaces and strong attractive forces between them. There is almost no room for the particles to be pushed closer, so the compressibility of solids is nearly zero.

2. Why Some Solids are Sublime in Nature

Some solids, like iodine, naphthalene and camphor, have very weak intermolecular forces between their particles. On heating, their surface particles gain enough energy to escape directly into the vapour state without melting first. Such solids are said to be sublime in nature. Sublimation is the direct conversion of a solid into gas (vapours) on heating, without passing through the liquid state.

3. Melting Point

Melting point is the fixed temperature at which a solid changes into liquid at atmospheric pressure. Crystalline solids have a sharp melting point, while amorphous solids melt over a range of temperature.

Types of Solids

LESSON 33

Solids are of two main types, crystalline and amorphous, according to the arrangement of their particles.

1. Crystalline and Amorphous Solids

Crystalline Solids	Amorphous Solids
Particles arranged in a definite, regular lattice	Particles randomly arranged
Sharp melting point	Melt over a range of temperature
Definite geometrical shapes, e.g. NaCl and alum	No definite shape, e.g. glass and rubber
Anisotropic behaviour	Isotropic behaviour

Alum is a crystalline solid because it has a definite geometrical shape, sharp melting point and an ordered lattice; glass, plastic and rubber are amorphous.

Properties of Crystalline Solids

LESSON 34

Because their particles are arranged in a regular order, crystalline solids show special properties such as anisotropy, and many elements show allotropy and polymorphism.

1. Anisotropy

Anisotropy is the property of a substance of showing different values of certain physical properties, such as electrical conduction, thermal expansion and refractive index, when measured in different directions. Crystalline solids show anisotropy because their particles are arranged in a regular order, and the arrangement is different along different directions. For example, graphite conducts electricity parallel to its layers but not perpendicular to them, so graphite is anisotropic. Amorphous solids are isotropic because their random arrangement is the same in all directions.

2. Allotropy

Allotropy is the phenomenon in which the same element exists in two or more different crystal structures (forms) in the same physical state, e.g. diamond and graphite.

3. Transition Temperature

The transition temperature is the temperature at which two allotropic forms of the same element co-exist in equilibrium; below and above it, one or the other form is more stable. For example, rhombic sulphur is stable below 96 degree C and monoclinic sulphur above it, so 96 degree C (369K) is the transition temperature of sulphur, and both forms co-exist there. Another example is tin: white (metallic) tin changes into grey (powdery) tin below 13.2 degree C, which is its transition temperature. At the transition temperature the two forms have equal stability.

4. Polymorphism and Its Relation to Allotropy

Polymorphism is the phenomenon in which a solid substance exists in more than one crystalline shape (form) under different conditions of temperature and pressure. Allotropy is the same phenomenon applied to an element: when an element exists in two or more crystalline forms, the forms are called allotropes. Thus allotropy is a special case of polymorphism for elements. For example, carbon shows allotropy as diamond and graphite; sulphur is polymorphic (allotropic) as rhombic and monoclinic sulphur. Compounds can also be polymorphic, e.g. calcium carbonate as calcite and aragonite. Isomorphism is different: it is the phenomenon in which different solid substances exist in the same crystalline structure, e.g. NaCl and MgO, which both crystallize in the same cubic (rock-salt) structure and so have the same geometrical shapes.

Types of Crystalline Solids

LESSON 35

Crystalline solids are classified into ionic, covalent, metallic and molecular types according to the particles that build the lattice and the forces between them.

1. Ionic Solids

- They are hard and brittle because of strong electrostatic forces between oppositely charged ions.
- They have high melting and boiling points.
- They do not conduct electricity in the solid state but conduct when molten or in aqueous solution, because the ions become free to move.
- They are soluble in polar solvents like water but insoluble in non-polar solvents.

In NaCl, each Na+ ion is surrounded octahedrally by six Cl- ions, and each Cl- ion by six Na+ ions (6:6 coordination).

2. Covalent Solids: Diamond and Graphite

Diamond is a covalent (network) solid: every carbon atom is joined to four others by covalent bonds in a giant network. Diamond is a non-conductor of electricity because each carbon atom uses all four of its valence electrons in covalent bonds, so there are no free electrons to carry current. In graphite, each carbon atom is bonded to only three others in flat layers, and the fourth electron of each atom is spare (delocalized) over the layer. These free electrons can move along the layers when a voltage is applied, so graphite conducts electricity, but only parallel to its layers, making it anisotropic.

3. Metallic Solids

Metallic solids consist of positive metal ions (kernels) arranged in a regular lattice, surrounded by a sea of free, delocalized valence electrons; the attraction between the kernels and this electron sea is the metallic bond. They are generally hard, lustrous, malleable and ductile, and have high melting points and densities. They conduct electricity because the free electrons can move through the lattice and carry charge when a voltage is applied. They conduct heat because these mobile electrons quickly transfer kinetic energy from the hot part to the cold part of the metal. Examples are iron, copper, aluminium and sodium.

4. Molecular Solids

Molecular solids consist of individual molecules held together only by weak intermolecular forces such as London forces, dipole-dipole attractions or hydrogen bonds, while the covalent bonds inside each molecule are strong. Very little energy is needed to overcome these weak forces, so molecular solids are soft and possess low melting points: ice (whose molecules are held by hydrogen bonds), solid CO2 (dry ice) and naphthalene are all soft solids with low melting or sublimation points, unlike ionic or covalent network solids.

Crystal Lattice

LESSON 36

The regular three-dimensional arrangement of particles in a crystal is described through its smallest repeating unit, the unit cell.

1. Unit Cell

A unit cell is the smallest portion (building block) of a crystal lattice which, when repeated over and over again in three dimensions, builds up the whole crystal. It is described by three axial distances (edge lengths) a, b and c, measured along the three axes, and three axial angles: alpha between the b and c axes, beta between the a and c axes, and gamma between the a and b axes. On the basis of these parameters, crystals are classified into seven crystal systems, such as cubic (a = b = c, all angles 90 degree) and orthorhombic (a, b and c all different in length but all angles 90 degree).

2. Number of Ions in the NaCl Unit Cell

In the NaCl unit cell, the Cl- ions occupy the 8 corners (8 x 1/8 = 1) and the 6 face centres (6 x 1/2 = 3), giving 4 Cl- ions. The Na+ ions occupy the 12 edges (12 x 1/4 = 3) and the body centre (1), giving 4 Na+ ions. Thus each unit cell of sodium chloride contains four Na+ and four Cl- ions, i.e. four formula units.

3. Lattice Energy

Lattice energy is the energy released when one mole of an ionic crystal is formed from its isolated gaseous ions; it is a measure of the strength of the ionic lattice. It increases when the size of the ions decreases, because smaller ions come closer together and attract each other more strongly, e.g. LiF has a higher lattice energy than NaBr. It also increases greatly with the charge on the ions, since the electrostatic attraction is proportional to the product of the charges; thus MgO (with +2 and -2 ions) has a much higher lattice energy than NaCl (with +1 and -1 ions).

Chemical Equilibrium

UNIT 7

Reversible Reactions and Dynamic Equilibrium

LESSON 37

Many reactions do not go to completion; instead they reach a balance point called equilibrium.

1. Reversible Reaction

A reversible reaction is a reaction which can proceed in both the forward and the backward direction under the same conditions, and is written with a double arrow; for example, N2 + 3H2 <=> 2NH3. Such reactions never go to completion because the products react to reform the reactants.

2. Chemical Equilibrium

Chemical equilibrium is the state of a reversible reaction at which the rate of the forward reaction becomes equal to the rate of the backward reaction. At equilibrium there is no net change in the concentrations of reactants and products with time, although both reactions continue at the microscopic level, so it is a dynamic equilibrium.

3. Law of Mass Action and Derivation of Kc

The law of mass action states that, at constant temperature, the rate of a chemical reaction is proportional to the active masses (molar concentrations) of the reacting species. Active mass means the molar concentration of a substance, i.e. number of moles per dm3 of solution (or partial pressure for a gas). For the general reaction aA + bB <=> cC + dD, the forward rate = kf [A]^a [B]^b and the backward rate = kb [C]^c [D]^d. At equilibrium the two rates are equal, so kf [A]^a [B]^b = kb [C]^c [D]^d. Rearranging gives kf/kb = [C]^c [D]^d / ([A]^a [B]^b) = Kc, the equilibrium constant. Its value depends only on temperature and is fixed for a given balanced equation.

4. Writing Kc and Kp Expressions

- 2NO2(g) + 7H2(g) <=> 2NH3(g) + 4H2O(g): Kc = [NH3]^2 [H2O]^4 / ([NO2]^2 [H2]^7), and Kp = (P NH3)^2 (P H2O)^4 / ((P NO2)^2 (P H2)^7).
- 2H2S(g) + 3O2(g) <=> 2H2O(g) + 2SO2(g): Kc = [H2O]^2 [SO2]^2 / ([H2S]^2 [O2]^3), and Kp = (P H2O)^2 (P SO2)^2 / ((P H2S)^2 (P O2)^3).

In each expression the product concentrations are written in the numerator and the reactant concentrations in the denominator, each raised to the power of its coefficient in the balanced equation. Kp uses partial pressures in place of concentrations for gaseous reactions. The two constants are related by Kp = Kc(RT)^Dn, where Dn is the change in gas moles: for PCl5(g) <=> PCl3(g) + Cl2(g), Dn = +1 so Kp > Kc, while for A(g) + B(g) <=> 2C(g), Dn = 0 so Kp = Kc.

5. Predicting Direction and Extent of Reaction from Kc

- **Direction:** the reaction quotient Qc is calculated with the actual concentrations; if Qc < Kc the reaction proceeds forward, if Qc > Kc it proceeds backward, and if Qc = Kc the system is at equilibrium.
- **Extent:** a large Kc (greater than 10^3) means products dominate and the reaction goes nearly to completion; a small Kc (less than 10^-3) means reactants dominate and the reaction hardly proceeds; an intermediate value means appreciable amounts of both are present.

6. Worked Numericals

Numerical 1: At 444 degree C the reaction of hydrogen and iodine is performed in a sealed 1 dm3 steel vessel: H2(g) + I2(g) <=> 2HI(g). The equilibrium mixture contains 1 mole of H2, 1 mole of I2 and 7 moles of HI. Calculate (a) Kc (b) the initial concentration of H2 and I2. Solution: The vessel is 1 dm3, so moles equal molar concentrations. (a) Kc = [HI]^2/([H2][I2]) = (7)^2/(1 x 1) = 49. (b) 7 moles of HI were formed from 3.5 moles of H2 and 3.5 moles of I2, so initial H2 = 1 + 3.5 = 4.5 moles and initial I2 = 4.5 moles.

Numerical 2: In an equilibrium study of H2(g) + I2(g) <=> 2HI(g) at 440 degree C, 1.2 mole of H2 and 1.2 moles of I2 were injected into an evacuated 1000 cm3 sealed flask. If Kc = 49, calculate the equilibrium concentrations. Solution: Flask volume = 1 dm3, so initial [H2] = [I2] = 1.2 mol/dm3. Let x mol/dm3 of H2 react; at equilibrium [H2] = [I2] = 1.2 - x and [HI] = 2x. Kc = (2x)^2/(1.2 - x)^2 = 49, so 2x/(1.2 - x) = 7, giving x = 0.933. Hence [H2] = [I2] = 0.267 mol/dm3 and [HI] = 1.866 mol/dm3.

Factors Affecting Equilibrium (Le-Chatelier's Principle)

LESSON 38

A system at equilibrium responds to any disturbance by shifting so as to cancel the disturbance.

1. Le-Chatelier's Principle

Le-Chatelier's principle states that if a stress (change in concentration, pressure or temperature) is imposed on a system at equilibrium, the system shifts in the direction that reduces the effect of the applied stress.

2. Effect of Pressure

A pressure increase shifts the equilibrium towards the side with fewer gas moles. The equilibrium of N2(g) + O2(g) <=> 2NO(g) is not affected by pressure because the number of gas moles is the same on both sides (2 = 2, Dn = 0); for this endothermic reaction only a temperature change shifts the equilibrium and changes Kc.

3. Effect of Catalyst

When a catalyst is added to a chemical system at equilibrium, Kc remains unchanged: a catalyst speeds up the forward and backward reactions equally and only helps equilibrium to be reached sooner.

4. Application: Increasing the Yield of SO3 in the Contact Process

For 2SO2(g) + O2(g) <=> 2SO3(g) (DH = -ve), the reaction is exothermic and shows a decrease in gas moles (3 moles give 2 moles). First, increase the pressure: the equilibrium shifts towards the side with fewer moles, producing more SO3. Second, decrease the temperature: for an exothermic reaction, lowering the temperature favours the forward direction, increasing SO3 (in practice a moderate 450 degree C is used with a V2O5 catalyst so the rate stays reasonable). Third, increase the concentration of SO2 or O2 (oxygen is used in excess), or remove SO3 as it is formed by absorbing it in concentrated H2SO4; both shift the equilibrium forward.

Industrial Application of Le-Chatelier's Principle (Haber's Process)

LESSON 39

The manufacture of ammonia is the classic industrial application of Le-Chatelier's principle.

1. The Haber Process Equilibrium

N2 + 3H2 <=> 2NH3, DH = -92 KJ/mol. There are 4 moles of gas on the left and 2 on the right, and the reaction is exothermic.

2. Choosing the Industrial Conditions

- **Pressure:** a high pressure (about 200 atm) shifts the equilibrium towards ammonia, the side with fewer gas moles.
- **Temperature:** a lower temperature favours ammonia because the reaction is exothermic, but would make the rate too slow, so a compromise temperature of about 450 degree C is used.
- **Catalyst:** an iron catalyst speeds up the attainment of equilibrium without changing Kc.
- **Removal of product:** ammonia is condensed and removed so the equilibrium keeps shifting forward.

Solubility Product

LESSON 40

Sparingly soluble salts dissolve only to a very small extent, and their saturated solutions are described by a special equilibrium constant, Ksp.

1. Solubility Product (Ksp)

The solubility product (Ksp) is the product of the molar concentrations of the ions of a sparingly soluble salt in its saturated solution at a given temperature, each concentration raised to the power of its coefficient in the dissociation equation.

2. Ksp Expressions

- BaF2 <=> Ba2+ + 2F-: Ksp = [Ba2+][F-]^2
- Li2C2O4 <=> 2Li+ + C2O4 2-: Ksp = [Li+]^2 [C2O4 2-]
- MgCO3 <=> Mg2+ + CO3 2-: Ksp = [Mg2+][CO3 2-]
- Ag3PO4 <=> 3Ag+ + PO4 3-: Ksp = [Ag+]^3 [PO4 3-]

3. Worked Numericals

Numerical 1: Lead fluoride (PbF2) is used in glass coating to reflect IR rays. Its solubility in water at 25 degree C is 0.58 g/dm3. Calculate its Ksp (atomic mass of Pb = 207 and F = 19). Solution: Molar mass of PbF2 = 207 + 2(19) = 245 g/mol. Molar solubility s = 0.58/245 = 2.37 x 10^-3 mol/dm3. PbF2 <=> Pb2+ + 2F-, so [Pb2+] = s and [F-] = 2s. Ksp = s(2s)^2 = 4s^3 = 4 x (2.37 x 10^-3)^3 = 5.3 x 10^-8 mol3/dm9.

Numerical 2: A solution is prepared by mixing 200 cm3 of 2.4 x 10^-4M Ca(NO3)2 and 300 cm3 of 4.5 x 10^-2M K2CO3. Will CaCO3 precipitate on cooling to 25 degree C if Ksp of CaCO3 at 25 degree C is 3.8 x 10^-9? Solution: Total volume = 500 cm3. After mixing, [Ca2+] = 2.4 x 10^-4 x (200/500) = 9.6 x 10^-5 M, and [CO3 2-] = 4.5 x 10^-2 x (300/500) = 2.7 x 10^-2 M. Ionic product Qsp = 9.6 x 10^-5 x 2.7 x 10^-2 = 2.6 x 10^-6. Since Qsp is greater than Ksp (3.8 x 10^-9), the solution is supersaturated and CaCO3 will precipitate.

Common Ion Effect

LESSON 41

Adding an ion that is already present in an equilibrium suppresses the dissolution or ionization of the electrolyte.

1. What is the Common Ion Effect?

The common ion effect is the decrease in the solubility (or ionization) of a sparingly soluble salt or weak electrolyte when a highly soluble electrolyte containing an ion common to it is added to its solution. It is an application of Le-Chatelier's principle.

2. Examples

- The solubility of AgCl decreases when NaCl is added, because the extra Cl- ions shift the equilibrium AgCl(s) <=> Ag+ + Cl- to the left and more AgCl precipitates.
- NH4Cl suppresses the ionization of weak NH4OH by supplying NH4+ ions.

3. Applications

- Qualitative analysis of salts.
- Salting out of soaps.
- Purification of salts.

Acids, Bases and Salts

UNIT 8

Acidic, Basic and Amphoteric Substances

LESSON 42

Oxides and other substances are classified as acidic, basic or amphoteric according to how they react with acids and bases.

1. Types of Oxides

- **Acidic oxides** react with bases, e.g. CO2.
- **Basic oxides** react with acids, e.g. K2O and CaO.
- **Amphoteric oxides** react with both acids and bases to form salt and water.

2. Amphoteric Behaviour of Al2O3

Al2O3 is an amphoteric oxide because it reacts with both acids and base: with HCl it behaves as a base and with NaOH it behaves as an acid, forming salt and water in each case. This dual behaviour is the test of an amphoteric substance.

3. Identifying Properties of Bases

- Bases have bitter tastes.
- They have high pH values.
- They react with acids to form salts.
- They turn red litmus blue (acids turn blue litmus red).

The Bronsted-Lowry Theory of Acids and Bases

LESSON 43

The Bronsted-Lowry theory defines acids and bases in terms of proton transfer, so it applies beyond aqueous solutions.

1. Definitions

- An **acid** is a species which tends to donate a proton (H+).
- A **base** is a species which tends to accept a proton.

2. Conjugate Acid-Base Pairs

When an acid donates a proton, the species left behind can accept a proton back and is called its conjugate base; similarly, a base accepting a proton becomes its conjugate acid. A conjugate acid-base pair is two species that differ by only one proton, one on each side of the equation. For example, in HCl + H2O <=> H3O+ + Cl-, HCl/Cl- and H3O+/H2O are conjugate pairs. Water can act as an acid with NH3 and as a base with HCl, so it is amphoteric in this theory.

Conjugate Acid-Base Pairs

LESSON 44

Every acid has its own conjugate base, and every base its own conjugate acid; members of a pair differ by exactly one proton (H+), e.g. NH4+ and NH3.

1. Conjugate Bases of Given Acids

A conjugate base is formed when an acid loses one proton:

- H2SO4 -> HSO4- (bisulphate ion)
- H2S -> HS- (bisulphide ion)
- NH4+ -> NH3 (ammonia)
- HCOOH -> HCOO- (formate ion)

In each pair the acid and its conjugate base differ by exactly one proton, and a strong acid has a weak conjugate base. The conjugate acid of NH3 is NH4+, formed by gaining a proton.

2. Identifying Conjugate Pairs in Reactions

- **Acetic acid and ammonia:** CH3COOH + NH3 <=> CH3COO- + NH4+. CH3COOH (acid) and CH3COO- (its conjugate base) form one pair, and NH4+ (conjugate acid) and NH3 (base) form the other pair; the proton passes from acetic acid to ammonia.
- **Hydrochloric acid and water:** HCl + H2O <=> Cl- + H3O+. HCl (acid) and Cl- (its conjugate base) form one pair, and H3O+ (conjugate acid) and H2O (base) form the other pair.

Strengths of Acids and Bases

LESSON 45

The strength of an acid or base depends on how completely it ionizes in water, and the pH scale measures the resulting acidity.

1. Strong and Weak Acids

Acid strength depends on the extent of ionization: H2SO4 is a stronger acid than CH3COOH because its degree of ionization is high — it ionizes almost completely in water — while CH3COOH ionizes only slightly. A strong acid has a weak conjugate base, and a weak acid has a strong conjugate base.

2. Self-Ionization of Water and Kw

Self-ionization of water is the process in which one water molecule donates a proton to another water molecule, producing hydronium and hydroxide ions: 2H2O <=> H3O+ + OH-. The ionic product of water is expressed as Kw = [H+][OH-] (more exactly [H3O+][OH-]). Its value at 25 degree C is 1 x 10^-14 mol2/dm6. In pure water [H+] = [OH-] = 1 x 10^-7 M, which is why pure water is neutral with pH 7 at this temperature.

3. pH and pOH

- pH is the negative logarithm of the hydrogen ion concentration, pH = -log[H+]; it measures the acidity of a solution on a scale from 0 to 14.
- pOH is the negative logarithm of the hydroxide ion concentration, pOH = -log[OH-].
- From Kw = [H+][OH-] = 1 x 10^-14 at 25 degree C, taking the negative log of both sides gives pH + pOH = 14.
- A solution with pH less than 7 is acidic, equal to 7 neutral, and greater than 7 basic.

4. Worked Numericals

Numerical 1: A solution is made by dissolving 14.8g HCl in water at 25 degree C. If the volume of solution is 750 cm3 and HCl is assumed to be completely ionized, calculate its pH. Solution: Moles of HCl = 14.8/36.5 = 0.405 mol in 0.750 dm3, so [H+] = 0.405/0.750 = 0.54 mol/dm3. pH = -log(0.54) = 0.27. (Working with the given data gives pH about 0.27; the book prints the answer as pH = 1.2, which corresponds to about one-tenth of this concentration.)

Numerical 2: The hydroxide ion concentration in an antiseptic solution at 25 degree C is 3.5 x 10^-4M. Calculate its pH. Solution: pOH = -log(3.5 x 10^-4) = 4 - log 3.5 = 4 - 0.54 = 3.46. Using pH + pOH = 14, pH = 14 - 3.46 = 10.54, so the solution is basic. (The book prints the answer as pH = 8; the value calculated from the given hydroxide ion concentration is about 10.5.)

Lewis Definition of Acids and Bases

LESSON 46

The Lewis theory is the most general theory of acids and bases because it is based on electron pairs rather than protons.

1. Lewis Definitions

- A Lewis **acid** is a substance that can accept an electron pair.
- A Lewis **base** is a substance that can donate an electron pair.

Thus BF3 is a Lewis acid because boron has an incomplete octet and accepts a lone pair from NH3, which is a Lewis base; the product is a coordinate (adduct) compound.

2. Advantages over the Bronsted-Lowry Theory

- It is not limited to proton transfer or to aqueous solutions, so reactions like BF3 + NH3 and metal ion complex formation are included as acid-base reactions.
- It explains the acidic behaviour of oxides such as CO2 and SO3 and of metal ions.
- Hence it is the most general theory of acids and bases.

Buffer Solutions and their Applications

LESSON 47

A buffer solution keeps its pH almost constant when small amounts of acid or base are added to it.

1. What is a Buffer Solution?

A buffer solution is a mixture of a weak acid with its salt of a strong base (acidic buffer), or a weak base with its salt of a strong acid (basic buffer), which resists a change in pH when a small amount of acid or base is added. Examples are CH3COOH/CH3COONa, Na2CO3/NaHCO3 and NH4OH/NH4Cl. NaOH/HCl is not a buffer because it is a mixture of a strong base and a strong acid, not a weak acid/base with its salt.

2. How a Buffer Resists Change of pH

In an acidic buffer like CH3COOH/CH3COONa, added H+ ions are removed by the acetate ions to form weakly ionized acetic acid, and added OH- ions are removed by the acetic acid to form acetate and water, so the pH stays nearly constant. The range over which a buffer is effective is called its buffer range.

3. Applications of Buffer Solutions

- Maintaining the pH of blood (about 7.4).
- Fermentation and dyeing industries.
- Chemical analysis.
- Medicines and shampoos.

Salts, their Types and Applications

LESSON 48

Salts are the products of neutralization, and their behaviour in water depends on the strength of the parent acid and base.

1. What is a Salt?

A salt is an ionic compound formed by the neutralization reaction of an acid and a base, in which the H+ of the acid is replaced by a metal or ammonium ion, e.g. HCl + NaOH -> NaCl + H2O.

2. Types of Salts

- **Acidic salts** still contain replaceable hydrogen and are formed by partial neutralization of a polybasic acid, e.g. NaHSO4 and NaHCO3.
- **Basic salts** still contain an OH group and are formed by partial neutralization of a polyacidic base, e.g. Mg(OH)Cl.
- **Neutral (normal) salts** are formed by complete neutralization and contain neither replaceable H nor OH, e.g. NaCl, Na2SO4 and CaCO3.

3. Hydrolysis of Salts

Hydrolysis is the reaction of the cation or anion of a salt with water, which changes the pH of the solution.

- **K2CO3** (strong base + weak acid): the CO3 2- ion hydrolyzes to produce OH- ions, so the solution is alkaline (pH > 7). Similarly Na2CO3 solution is alkaline because CO3 2- + H2O -> HCO3- + OH-.
- **(NH4)2SO4** (weak base + strong acid): the NH4+ ion hydrolyzes to produce H+ ions, so the solution is acidic (pH < 7). Similarly NH4Cl solution is acidic because NH4+ + H2O -> NH4OH + H+.
- **NaNO3** (strong base + strong acid): neither Na+ nor NO3- hydrolyzes appreciably, so the solution remains neutral (pH = 7). NaCl likewise does not hydrolyze and does not alter the pH of a solution.
- Na2CO3 is the salt formed by neutralization of a weak acid (H2CO3) and a strong base (NaOH), and NH4CN is the salt of a weak acid (HCN) and a weak base (NH4OH).

In general, cation hydrolysis gives an acidic solution and anion hydrolysis gives a basic solution.

Chemical Kinetics

UNIT 9

Chemical Kinetics

LESSON 49

Chemical kinetics is the study of the rates of chemical reactions and the factors that affect them. Some reactions are fast, some moderate and some slow; ionic reactions, which involve only the combination of ions already present in solution and need almost no bond breaking, are fast (instantaneous) reactions.

1. Rate of Reaction

Rate of reaction is the decrease in concentration of a reactant or the increase in concentration of a product per unit time during a chemical reaction. For A -> B, the rate of disappearance of A is written -dA/dt (with a negative sign because the concentration of A decreases, so that the rate is a positive quantity) and the rate of appearance of B is +dB/dt.

2. Velocity of Reaction

Velocity of reaction is the rate at a particular instant (instantaneous rate), measured from the slope of a concentration-time graph at that point. The best alternative term for the velocity of reaction is the instantaneous rate.

3. Order of Reaction

Order of reaction is the sum of the powers to which the concentrations of the reactants are raised in the experimental rate law; it may be a whole number, a fraction or even zero. R = K means the rate does not depend on any concentration, so the reaction is of zero order.

4. Rate Constant

The rate constant (specific rate constant) is the rate of reaction when the molar concentration of each reactant is unity; its value depends only on temperature (and a catalyst), not on concentration or time.

Rate of Reaction	Rate Constant
Change in concentration per unit time	Rate when each concentration is unity
Changes as the reaction proceeds	Stays constant at a fixed temperature

5. Elementary and Overall Reactions

An elementary reaction takes place in a single step, while an overall reaction is the sum of several elementary steps (its mechanism).

Rate of Reaction

LESSON 50

The rate of a reaction is measured experimentally, and the results are summarized in a rate law that shows how the rate depends on concentrations.

1. Factors Influencing the Rate of Reaction

The rate of a chemical reaction is influenced by the nature of the reactants, concentration of reactants, surface area of solid reactants, temperature, light (radiations), molecular size and the presence of a catalyst.

2. Effect of Concentration

Increasing the concentration puts more particles in the same volume, so collisions become more frequent and the rate increases; reactants in solution react faster at high concentration. For many reactions the rate is proportional to a power of the concentration, as shown by the rate law.

3. Effect of Surface Area

For a solid reactant, breaking it into small pieces or powder exposes more particles at the surface where reaction occurs. Hence powdered marble ($CaCO_3$) reacts quickly with hydrochloric acid compared with a solid lump of marble, and coal dust burns faster than coal lumps.

4. Everyday Effects of Temperature and Other Factors

- Milk sours more rapidly in summer than in winter because in summer the temperature is higher, so the bacteria that sour milk work faster and the reactions proceed at a greater rate.
- Raising the temperature increases the kinetic energy of molecules, so more of them can cross the activation energy barrier and the number of effective collisions rises sharply.

5. Rate Law and Order

For the rate law of 2NO + Br2 -> 2NOBr, Rate = K[NO]^2[Br2]: the reaction is second order with respect to NO (power 2) and first order with respect to Br2 (power 1), so the overall order is 2 + 1 = 3. Order is found experimentally from the rate law, not simply from the balancing coefficients, although here they agree. Similarly, if for x + y -> z doubling x makes the rate four times and doubling y doubles the rate, the rate law is R = K[x]^2[y]^1.

6. Units of the Rate Constant

Order	Units of K
Zero order	mol dm^-3 s^-1 (same units as rate)
First order	s^-1
Second order	dm^3 mol^-1 s^-1

7. Worked Numericals

Numerical 1: Decomposition of NO2 into NO and O2 is a second order reaction: 2NO2 -> 2NO + O2. If the rate constant at a certain temperature is 3.8 x 10^-4 dm3 mol^-1 s^-1 and the initial concentration of NO2 is 0.38M, calculate the initial rate. Solution: Rate = K[NO2]^2 = 3.8 x 10^-4 x (0.38)^2 = 3.8 x 10^-4 x 0.1444 = 5.48 x 10^-5 mol dm^-3 s^-1.

Numerical 2: The overall rate law for A + B -> C is R = K[A][B]. If the initial concentrations of A and B were 0.43M and 0.78M while the initial rate was 3.8 x 10^-3 M s^-1, determine the rate constant and its unit. Solution: K = Rate/([A][B]) = 3.8 x 10^-3/(0.43 x 0.78) = 3.8 x 10^-3/0.3354 = 1.13 x 10^-2 dm3 mol^-1 s^-1 (second order overall).

Numerical 3: The reaction 2NO + Cl2 -> 2NOCl was studied at 25 degree C: Experiment 1: [NO] = 0.1, [Cl2] = 0.1, rate = 2.52 x 10^-3; Experiment 2: [NO] = 0.1, [Cl2] = 0.2, rate = 5.04 x 10^-3; Experiment 3: [NO] = 0.2, [Cl2] = 0.1, rate = 10.05 x 10^-3. Find the rate law and order. Solution: Comparing experiments 1 and 2, [Cl2] doubles and the rate doubles, so the reaction is first order in Cl2. Comparing experiments 1 and 3, [NO] doubles and the rate becomes about four times, so the reaction is second order in NO. Rate law: Rate = K[NO]^2[Cl2]; overall order = 3.

Collision Theory, Transition State and Activation Energy

LESSON 51

Collision theory explains reaction rates in terms of the collisions between molecules.

1. Collision Theory

A reaction occurs only when molecules collide with sufficient energy (at least the activation energy) and with proper orientation; such collisions are called effective collisions.

2. How the Rate Increases with Temperature

When the temperature rises, the average kinetic energy of the molecules increases and they move faster, so the total number of collisions per second increases. More importantly, the fraction of molecules having energy equal to or greater than the activation energy increases greatly. As a result, the number of effective collisions rises sharply and the reaction rate increases. This is why a rise of about 10 degree C often roughly doubles the rate of a reaction.

3. Activation Energy and Activated Complex

Activation energy is the additional (minimum) energy which the reacting molecules must possess, above their average kinetic energy, for their collision to be effective and produce products. As reactant molecules approach, their energy rises to a maximum, forming a high-energy, unstable and short-lived state called the activated complex (transition state), which then breaks down into products with release of energy. On a potential energy diagram, energy is plotted against the progress of the reaction: the curve rises from the reactants' level to a peak (the activated complex) and then falls to the products' level. The height of the peak above the reactants is the activation energy, and the difference between the product and reactant levels is the enthalpy change of the reaction.

Catalysis

LESSON 52

A catalyst changes the rate of a reaction without being consumed in it.

1. How a Catalyst Works

The purpose of using a catalyst in a chemical reaction is to change the activation energy: a catalyst provides an alternative path of lower activation energy, so the reaction becomes faster without changing the equilibrium constant or the enthalpy of reaction.

2. Homogeneous and Heterogeneous Catalysts

- A **homogeneous catalyst** is in the same phase as the reactants, e.g. an acid catalyst in solution.
- A **heterogeneous catalyst** is in a different phase, e.g. solid iron in Haber's process with gaseous reactants.

3. Positive Catalysts and Inhibitors

A positive catalyst increases the rate by lowering the activation energy, while an inhibitor (negative catalyst) decreases the rate. For example, glycerin acts as a negative catalyst (inhibitor) and slows the decomposition of H_2O_2, whereas MnO_2 speeds it up.

Solution & Colloids

UNIT 10

General Properties of Solution

LESSON 53

A solution is a homogeneous mixture of a solute in a solvent. Whether two liquids form a solution, and how much gas dissolves, depend on the nature of the substances and on pressure.

1. Miscible and Immiscible Liquids

Miscible liquids are liquids which mix completely in all proportions to form a homogeneous solution, e.g. water and ethanol. Immiscible liquids do not mix and separate into two layers, e.g. oil and water, and water and benzene form a completely immiscible pair. n-Hexane (petrol) is immiscible in water because n-hexane is a non-polar liquid whose molecules are held by weak London forces, whereas water is polar and strongly hydrogen bonded; the water molecules attract each other much more strongly than they attract hexane molecules, so hexane cannot break into the water structure — following the rule "like dissolves like", the two liquids remain immiscible.

2. Effect of Pressure on Solubility

Pressure plays a significant role in the solubility of a gas in a liquid. The solubility of oxygen in water increases with pressure (Henry's law): when the pressure of the gas above the water is increased, more gas molecules strike the surface and are forced into solution. The solubility of glucose in water, however, has a negligible effect of pressure, because glucose is a solid solute and solids and liquids are almost incompressible, so changing the external pressure does not change the closeness of its particles or its dissolving equilibrium to any noticeable extent.

Concentration Units

LESSON 54

The concentration of a solution tells how much solute is dissolved in a given amount of solvent or solution, and it can be expressed in several units.

1. Units of Concentration

- **Percentage composition:** weight/weight, weight/volume and volume/volume. A 15% W/W KOH solution is prepared by mixing 15g KOH in 85g water (15g solute in 100g solution).
- **Molarity (M):** moles of solute per dm3 of solution. A 1 molar solution in 2 dm3 contains 1 x 2 = 2 moles of solute.
- **Molality (m):** moles of solute per kg of solvent.
- **Mole fraction (X):** ratio of the number of moles of a component to the total number of moles of all the components.

2. Molarity and Molality

Molarity is the number of moles of solute dissolved in 1 dm3 (litre) of solution. Molality is the number of moles of solute dissolved in 1 kg of solvent. Molarity depends on temperature because the volume of the solution expands or contracts with temperature, while molality does not depend on temperature because it is based on the mass of the solvent, which does not change with temperature. For this reason molality is preferred in work involving colligative properties.

3. Mole Fraction

Mole fraction is the ratio of the number of moles of a component to the total number of moles of all the components in the solution. For a component A, XA = nA/(nA + nB). It has no units, each mole fraction lies between 0 and 1, and the sum of the mole fractions of all components is always equal to 1.0.

4. Worked Numericals

Numerical 1: Automotive antifreeze is a 60% (w/w) aqueous solution of ethylene glycol (C2H6O2). Determine (a) molality of the solution (b) mole fraction of ethylene glycol. Solution: Take 100g of solution: it contains 60g ethylene glycol and 40g water. Molar mass of C2H6O2 = 62 g/mol, so moles of glycol = 60/62 = 0.968 mol. (a) Molality = 0.968/0.040 = 24.19 mol/kg. (b) Moles of water = 40/18 = 2.22 mol. Mole fraction of glycol X = 0.968/(0.968 + 2.22) = 0.3.

Numerical 2: A solution is prepared by mixing 46g ethanol (C2H5OH) and 180g water. Calculate the mole fraction of both components. Solution: Moles of ethanol = 46/46 = 1 mol. Moles of water = 180/18 = 10 mol. Total moles = 11. Mole fraction of ethanol X2 = 1/11 = 0.090, and mole fraction of water X1 = 10/11 = 0.909. The sum of the two mole fractions is 1, as required.

Raoult's Law

LESSON 55

Raoult's law relates the vapour pressure of a solution to its composition.

1. Statement and Mathematical Expression

Raoult's law states that the vapour pressure of a solution of a non-volatile solute is directly proportional to the mole fraction of the solvent in the solution. Consider a solution with n1 moles of solvent and n2 moles of solute; the mole fraction of solvent is X1 = n1/(n1 + n2). If P0 is the vapour pressure of the pure solvent and P the vapour pressure of the solution, then by Raoult's law P = P0 X1 = P0 n1/(n1 + n2). Rearranging, the relative lowering of vapour pressure is (P0 - P)/P0 = n2/(n1 + n2) = X2, the mole fraction of the solute. This expression is used to find the molar mass of the solute from the measured lowering of vapour pressure.

2. Why the Vapour Pressure is Lowered (Particle Basis)

In a pure solvent, the whole surface consists of solvent particles which escape into the vapour, producing its vapour pressure. When a non-volatile solute is dissolved, some surface positions are occupied by solute particles, which cannot escape into the vapour. Hence fewer solvent particles can leave the surface per unit time, and the rate of evaporation decreases. The vapour pressure of the solution therefore becomes lower than that of the pure solvent. This lowering depends on the fraction of solute particles, i.e. on the mole fraction of the solute.

3. Ideal and Non-Ideal Solutions

An ideal solution is one which perfectly obeys Raoult's law at all temperatures and concentrations.

Ideal Solution	Non-Ideal Solution
No heat is absorbed or evolved on mixing (DH mixing = 0)	Heat is absorbed or evolved on mixing
No change in volume on mixing (DV = 0)	May expand or contract on mixing
Forces between unlike molecules are the same as between like molecules	Forces between unlike molecules are different
Examples: benzene + toluene, n-hexane + n-heptane, methanol + ethanol	Example: ethanol + water; ethanol + acetone also does not obey Raoult's law

Colligative Properties of Solution

LESSON 56

Colligative properties are physical properties of a dilute solution which depend only on the number of solute particles (molecules or ions) present and not on their nature. The main colligative properties are lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure. Atmospheric pressure is not a colligative property.

1. Lowering of Vapour Pressure

Adding a non-volatile solute lowers the vapour pressure of the solvent because some surface positions are taken by non-volatile solute particles; by Raoult's law the relative lowering equals the mole fraction of the solute.

2. Elevation of Boiling Point

A liquid boils when its vapour pressure becomes equal to the atmospheric pressure. When a non-volatile solute is added, the vapour pressure is lowered, so the solution must be heated to a higher temperature to reach that pressure. Hence the boiling point is raised (elevated).

3. Osmosis and Osmotic Pressure

Osmosis is the spontaneous flow of solvent particles from a dilute solution (higher solvent concentration) to a concentrated solution through a semipermeable membrane. Osmotic pressure is the hydrostatic pressure which must be applied to the solution to stop this flow of solvent across the membrane; it is a colligative property.

4. Daily Life Examples of Osmosis

- Absorption of water by plant roots from the soil.
- Shrinking of raw mangoes and grapes in strong salt or sugar solution.
- Swelling of dried raisins placed in water.
- Use of salt to preserve pickles and meat by drawing water out of micro-organisms.

Reverse osmosis, used in R.O. plants, is based on applying pressure greater than the osmotic pressure.

Colloids

LESSON 57

Between true solutions and coarse suspensions lies the colloidal state, in which the particles are bigger than molecules but too small to settle.

1. True Solution, Colloidal Solution and Suspension

Property	True Solution	Colloidal Solution	Suspension
Particle size	Less than 1 nm (ions or small molecules)	Between 1 nm and 1000 nm	Larger than 1000 nm
Visibility	Particles cannot be seen even with a powerful microscope	Particles cannot be seen with an ordinary microscope but can be detected with an ultramicroscope or by the Tyndall effect	Particles can be seen with the naked eye or an ordinary microscope and settle down on standing
Appearance	Homogeneous; no Tyndall effect	Translucent; shows the Tyndall effect and Brownian movement	Opaque and heterogeneous

2. Types of Colloids

- A colloidal solution of a liquid dispersed in another liquid is called an **emulsion**, e.g. milk.
- A **gel** is liquid dispersed in a solid.
- A **sol** is a solid dispersed in a liquid.
- A **foam** is a gas dispersed in a liquid or solid.

Thermochemistry

UNIT II

Thermodynamics

LESSON 58

Thermodynamics deals with the energy changes that accompany physical and chemical processes. Its basic terms describe what is being studied and how its state is specified.

1. System and Surroundings

The system is the specific part of the universe under experimental study, and everything outside it is the surrounding; the two are separated by a boundary. An open system exchanges matter and energy with the surroundings, a closed system exchanges energy only, and an isolated system exchanges neither matter nor energy (heat), so heat transfer across the boundary of an isolated system is not feasible.

2. State and State Function

The state of a system is its condition described by measurable properties like temperature, pressure and volume. A state function is a property whose value depends only on the initial and final states and not on the path, e.g. internal energy, enthalpy and volume. Heat (q) and work (w) are path functions.

3. Internal Energy and Enthalpy

- **Internal energy (E)** is the sum of all kinetic and potential energies of the particles of a system.
- **Enthalpy (H)** is the total heat content of a system at constant pressure, defined as H = E + PV.

4. Internal Energy Change and Enthalpy Change

Internal energy change DE is the difference in internal energy between the final and initial states of a system; at constant volume it is equal to the heat absorbed, DE = qv. Enthalpy change DH is the heat change at constant pressure, DH = qp, and is related by DH = DE + PDV (or DH = DE + Dn(g) RT). DE and DH are equal when there is no change in volume, i.e. when the reaction is carried out in a closed vessel at constant volume, or when there is no change in the number of gas moles (Dn(g) = 0). They are also nearly equal for reactions involving only solids and liquids, where DV is negligible. Entropy (disorder) is least in the most ordered state: ice at 0 degree C has the least entropy among the states of water.

Thermochemical Reactions

LESSON 59

Thermochemical reactions are chemical reactions accompanied by definite enthalpy changes, written together with their DH values.

1. Exothermic and Endothermic Reactions

- In an **exothermic reaction** heat is evolved to the surroundings, the products have less enthalpy than the reactants, and DH is negative; burning of fuel and neutralization are examples. Combustion is always exothermic, so the enthalpy of combustion is always negative.
- In an **endothermic reaction** heat is absorbed from the surroundings, the products have more enthalpy than the reactants, and DH is positive; decomposition of CaCO3 (lime) and photosynthesis are examples, as is the cracking of alkanes.

On an energy diagram, enthalpy is plotted against the progress of the reaction, with a peak for the activated complex in between. For an exothermic reaction the product level lies below the reactant level, while for an endothermic reaction the product level lies above it; the vertical difference between the two levels is DH in each case.

2. Standard Enthalpy of Formation

The standard enthalpy of formation (DHf degree) is the heat evolved or absorbed when one mole of a substance is produced from its constituent elements in their standard states, i.e. at 25 degree C (298K) and 1 atm pressure; by convention it is zero for elements in their most stable form. Thus the standard enthalpy of formation is zero for C(graphite), O2 and N2, but not for C(diamond), because graphite is the stable form of carbon.

3. Standard Enthalpy of Reaction

The standard enthalpy of reaction (DH degree) is the enthalpy change of a chemical reaction when all reactants and products are in their standard states. It is calculated as:

DH degree = Sum of DHf degree of products - Sum of DHf degree of reactants

These values allow the heat change of a reaction to be found without doing the experiment. The process Li+(g) + Cl-(g) -> LiCl(s) represents the lattice energy, the energy released when one mole of an ionic solid forms from its gaseous ions.

4. Worked Numericals

Numerical 1: In the Ostwald process, 4NH3(g) + 5O2(g) -> 4NO(g) + 6H2O(l). Determine the standard heat of reaction from DHf degree: NH3(g) = -46.19 KJ/mol, NO(g) = 90.25 KJ/mol, H2O(l) = -285.8 KJ/mol. Solution: Products: 4(90.25) + 6(-285.8) = 361 - 1714.8 = -1353.8 KJ. Reactants: 4(-46.19) + 5(0) = -184.76 KJ. DH degree = -1353.8 - (-184.76) = -1169.04 KJ for the reaction as written. The negative sign confirms that the oxidation of ammonia is strongly exothermic.

Numerical 2: Find the standard heat of combustion of iso-octane: C8H18(l) + 12.5 O2(g) -> 8CO2(g) + 9H2O(l), given DHf degree of CO2 = -393.5 KJ/mol, H2O = -285.8 KJ/mol and C8H18 = -223.8 KJ/mol. Solution: Products: 8(-393.5) + 9(-285.8) = -3148 - 2572.2 = -5720.2 KJ. Reactants: -223.8 + 0 = -223.8 KJ. DHc degree = -5720.2 - (-223.8) = -5496.4 KJ/mol. Burning one mole of iso-octane releases 5496.4 KJ of heat, which is why it is an efficient fuel.

First Law of Thermodynamics

LESSON 60

The first law of thermodynamics is the law of conservation of energy applied to chemical systems.

1. Statement of the First Law

Energy can neither be created nor destroyed; the total energy of a system and its surroundings remains constant. Mathematically:

DE = q + w

where DE is the change in internal energy, q the heat absorbed by the system and w the work done on the system. DE is a state function and depends only on the initial and final states, whereas q and w are path functions.

2. Pressure-Volume Work

Suppose a gas in a cylinder expands by DV against a constant external pressure P. The work done by the gas = force x distance = P x DV, so the work done on the system is:

w = -P DV

The negative sign shows that in expansion (DV positive) work is done by the system and its internal energy decreases, while in compression work is done on the system. At constant volume DV = 0, so no work is done in a thermochemical process kept at constant volume.

3. Applications at Constant Pressure and Constant Volume

- **At constant pressure:** the heat absorbed is used partly to increase internal energy and partly to do expansion work; this heat change is the enthalpy change, qp = DH = DE + PDV. Most laboratory and industrial reactions occur at constant (atmospheric) pressure, so enthalpy changes measured in an open apparatus apply the first law in this form. For a constant pressure process, DH = DE + PDV.
- **At constant volume:** DV = 0, so no pressure-volume work is done (w = 0) and all the heat absorbed appears as internal energy change, qv = DE. This is the basis of the bomb calorimeter, in which the heat of combustion of a fuel or food is measured at constant volume and DE is found directly.

4. Worked Numerical

Numerical: A thermochemical process is carried out at constant pressure of 8.52 atm. It absorbs 15.4 KJ energy from the surroundings, due to which an expansion in volume of 4.7 dm3 occurs. Calculate its change in internal energy. Solution: Work done by the system in expansion: w = -P DV = -8.52 x 4.7 = -40.04 dm3.atm. Converting (1 dm3.atm = 0.1013 KJ): w = -40.04 x 0.1013 = -4.06 KJ. Heat absorbed q = +15.4 KJ. By the first law, DE = q + w = 15.4 - 4.06 = 11.34 KJ. So the internal energy of the system increases by 11.34 KJ.

Hess's Law: Enthalpy Change Calculations

LESSON 61

Hess's law allows the enthalpy change of a reaction to be calculated from the enthalpy changes of other reactions.

1. Statement of Hess's Law

Hess's law states that if a reaction is carried out through several steps, the total enthalpy change for the overall reaction is equal to the sum of the enthalpy changes of the individual steps, whatever path is followed. It is a consequence of the first law of thermodynamics and of enthalpy being a state function. In cyclic form it is expressed as the sum of DH around a complete cycle being zero.

2. Applications of Hess's Law

- Calculating enthalpies of formation of compounds which cannot be measured directly (e.g. methane).
- Determining the enthalpy changes of slow or incomplete reactions.
- Constructing Born-Haber cycles to find the lattice energy of ionic compounds.
- Calculating the calorific values of fuels and foods.

3. Worked Numericals

Numerical 1: Calculate the standard enthalpy change for: (a) 2H2S(g) + 3O2(g) -> 2H2O(l) + 2SO2(g) (b) Fe2O3(s) + 3CO(g) -> 2Fe(s) + 3CO2(g), using DHf degree values (KJ/mol): H2S(g) = -20.6, H2O(l) = -285.8, SO2(g) = -296.8, Fe2O3(s) = -822.2, CO(g) = -110.5, CO2(g) = -393.5, elements = 0. Solution: (a) Products = 2(-285.8) + 2(-296.8) = -1165.2; reactants = 2(-20.6) + 3(0) = -41.2; DH degree = -1165.2 - (-41.2) = -1124 KJ. (b) Products = 2(0) + 3(-393.5) = -1180.5; reactants = -822.2 + 3(-110.5) = -1153.7; DH degree = -1180.5 - (-1153.7) = -26.8 KJ. Both reactions are exothermic.

Numerical 2: Calculate the standard enthalpy of formation of glycerol, C3H8O3, from: C(s) + O2(g) -> CO2(g) (DH = -393.5 KJ/mol); H2(g) + (1/2)O2(g) -> H2O(l) (DH = -285.8 KJ/mol); C3H8O3(l) + (7/2)O2(g) -> 3CO2(g) + 4H2O(l) (DH = -1654.1 KJ/mol). Solution: Combine 3 x (formation of CO2) + 4 x (formation of H2O) and subtract the combustion of glycerol: 3C + 3O2 -> 3CO2 gives 3(-393.5) = -1180.5 KJ; 4H2 + 2O2 -> 4H2O gives 4(-285.8) = -1143.2 KJ; reversing the combustion gives +1654.1 KJ. Adding all steps, the CO2, H2O and extra O2 cancel, leaving 3C(s) + 4H2(g) + (3/2)O2(g) -> C3H8O3(l). DHf degree = -1180.5 - 1143.2 + 1654.1 = -669.6 KJ/mol.

Numerical 3: Determine the lattice energy of rubidium chloride (RbCl) from the Born-Haber cycle using (KJ/mol): ionization potential of Rb = 403, electron affinity of Cl = -349, bond energy of Cl2 = 242, sublimation energy of Rb = 86.5, heat of formation of RbCl(s) = -430.5. Solution: In the Born-Haber cycle RbCl(s) is formed from its elements by two routes; by Hess's law both routes have the same total enthalpy. Steps: sublimation of Rb = +86.5; ionization of Rb = +403; dissociation of half a mole of Cl2 = (1/2)(242) = +121; electron affinity of Cl = -349; lattice energy = U. Sum of steps = enthalpy of formation: 86.5 + 403 + 121 - 349 + U = -430.5, so 261.5 + U = -430.5 and U = -692 KJ/mol. The negative sign shows that 692 KJ/mol is released when one mole of RbCl crystal forms from gaseous Rb+ and Cl- ions. The cycle steps are sublimation, ionization, dissociation, electron affinity and lattice formation.

Electrochemistry

UNIT 12

Oxidation-Reduction Concepts

LESSON 62

Electrochemistry rests on redox reactions, in which electrons are transferred from one species to another.

1. Oxidation and Reduction

- **Oxidation** is the loss of electrons (increase in oxidation number), e.g. Zn -> Zn+2 + 2e- shows oxidation.
- **Reduction** is the gain of electrons (decrease in oxidation number).
- **Redox reactions** are reactions involving transfer of electrons, in which oxidation and reduction take place simultaneously.
- An **oxidizing agent** oxidizes another substance and itself gets reduced, while a **reducing agent** reduces another substance and itself gets oxidized.
- During electrolysis, the reaction that takes place at the anode is always oxidation.

2. Oxidation Number

Oxidation number (oxidation state) is the charge, real or apparent, which an atom of an element has in a compound or ion, calculated by assuming that shared electrons belong completely to the more electronegative atom.

- Cr in H2CrO4: 2(+1) + x + 4(-2) = 0, so x = +6.
- S in K2S4O6: 2(+1) + 4x + 6(-2) = 0, so 4x = 10 and x = +2.5 (average).
- Fe in Fe3O4: 3x + 4(-2) = 0, so x = +8/3 (average).
- Cl in HClO4: +1 + x + 4(-2) = 0, so x = +7.
- Cr in Na2Cr2O7: 2(+1) + 2x + 7(-2) = 0, so 2x = 12 and x = +6.

3. Balancing Redox Equations by the Ion-Electron Method

By the ion-electron method each equation is split into oxidation and reduction half-reactions, each half is balanced for atoms and charge using H+, OH- and H2O as the medium requires, the electrons are equalized, and the halves are added.

- Cr2O7 2- + I2 (acidic): oxidation I2 + 6H2O -> 2IO3- + 12H+ + 10e-; reduction Cr2O7 2- + 14H+ + 6e- -> 2Cr3+ + 7H2O; overall: 5Cr2O7 2- + 3I2 + 34H+ -> 10Cr3+ + 6IO3- + 17H2O.
- MnO4- + Cl- (acidic): 2MnO4- + 10Cl- + 16H+ -> 2Mn2+ + 5Cl2 + 8H2O.
- Cl2 + OH- (basic): 3Cl2 + 6OH- -> 5Cl- + ClO3- + 3H2O.
- MnO4- + C2O4 2- (acidic): 2MnO4- + 5C2O4 2- + 16H+ -> 2Mn2+ + 10CO2 + 8H2O.
- Fe2+ + Cr2O7 2- (acidic): 6Fe2+ + Cr2O7 2- + 14H+ -> 6Fe3+ + 2Cr3+ + 7H2O.
- MnO4- + SO3 2- (basic): in basic medium permanganate is reduced to MnO2: 2MnO4- + 3SO3 2- + H2O -> 2MnO2 + 3SO4 2- + 2OH- (the book prints Mn+2 as the manganese product, but Mn+2 is not stable in basic medium).

Electrode, Electrode Potential and Electrochemical Series

LESSON 63

When a metal is dipped in a solution of its own ions, a potential difference is set up at the electrode, and all such potentials are compared on one scale.

1. Electrode Potential

Electrode potential is the potential difference set up between a metal electrode and the solution of its own ions when the metal is dipped into that solution; it arises because metal atoms tend to lose electrons (oxidation potential) or ions tend to gain them (reduction potential). The standard electrode potential (E degree) is the potential difference created between a metal electrode and a one molar solution of its ions at 25 degree C and 1 atmospheric pressure, measured against the standard hydrogen electrode.

2. Standard Hydrogen Electrode (SHE)

The standard hydrogen electrode is the reference electrode, assigned a potential of 0.0 volt. It consists of a platinum foil coated with platinum black, immersed in a 1M acid solution, with pure hydrogen gas at 1 atm pressure bubbling over it at 25 degree C. All other electrode potentials are measured relative to it, and half-cell reactions are written as reductions by convention.

3. Determination of the Electrode Potential of Copper

A Cu rod is dipped in 1M CuSO4 solution to form a half cell, which is connected through a salt bridge to a SHE and through wires to a voltmeter. The measured EMF of the cell Pt/H2/H+ // Cu+2/Cu is +0.34 V, and since the SHE potential is zero, the standard electrode potential of copper is E degree = +0.34 V, the positive sign showing copper is below hydrogen in the electrochemical series. In a Zn-SHE voltaic cell the anode half reaction is Zn -> Zn+2 + 2e-.

4. Electrochemical Series

The electrochemical series (ECS) is an arrangement of elements (electrodes) in order of their increasing standard electrode (reduction) potential values, with hydrogen in the middle as reference.

- Elements above hydrogen have negative E degree values and are strong reducing agents, the strongest being Li at the top.
- Elements below hydrogen have positive values and their ions are stronger oxidizing agents; F2, at the bottom of the series with the highest positive reduction potential, is the strongest oxidizing agent, while Li is the strongest reducing agent.
- A metal higher in the series can displace a metal lower in the series from its salt solution.
- The series is used to predict the products of electrolysis, the feasibility of a redox reaction and the EMF of a galvanic cell.
- The conduction of electricity through an electrolytic solution is due to the flow of ions, not electrons.

Batteries

LESSON 64

Batteries are galvanic cells that supply electrical energy from spontaneous redox reactions; they are of two kinds, primary and secondary.

1. Primary and Secondary Cells

Primary Cell	Secondary Cell
Cannot be recharged; once its chemicals are consumed it stops working and is thrown away	Can be recharged by passing electric current in the reverse direction, so it can be used again and again
Example: dry cell	Example: lead storage battery

The lead storage battery is a secondary (rechargeable) battery; its anode is made up of lead and its cathode of lead oxide, and it can be recharged.

2. The Dry Cell

The dry cell has an outer zinc container which serves as the anode, a central graphite rod as the cathode, and a paste of MnO2, NH4Cl, ZnCl2 and carbon powder around it acting as the electrolyte. At the anode Zn is oxidized to Zn+2 ions and releases electrons; at the cathode MnO2 is reduced while NH4+ ions take electrons. The electrons flow through the external circuit and light a bulb or run an appliance. Its EMF is about 1.5 V, and it leaks and runs down with time because the zinc container is gradually consumed.

3. The Copper-Zinc Galvanic Cell

The cell consists of a zinc rod dipped in ZnSO4 solution (anode half cell) and a copper rod dipped in CuSO4 solution (cathode half cell), connected externally by a wire through a voltmeter and internally by a salt bridge which allows ion flow but prevents mixing of the solutions. At the anode oxidation occurs: Zn -> Zn+2 + 2e-. At the cathode reduction occurs: Cu+2 + 2e- -> Cu. The overall cell reaction is Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s), with a standard EMF of 1.10 V. Electrons flow through the external circuit from the zinc anode (negative terminal) to the copper cathode (positive terminal), while current flows in the opposite direction. In the cell the Zn electrode loses mass and the Cu electrode gains mass.

4. Fuel Cell

A fuel cell is a typical galvanic cell which generates electricity from the reaction between hydrogen and oxygen: 2H2(g) + O2(g) -> 2H2O(l), with a continuous supply of hydrogen and oxygen.

Corrosion and its Prevention

LESSON 65

Corrosion is the slow eating away of a metal by its surroundings, and it is itself an electrochemical process.

1. What is Corrosion and What Causes It?

Corrosion is an electrochemical process in which atoms on the surface of a metal are oxidized (converted into harmful oxides) due to the action of the surrounding medium, such as moist air; the common example is the rusting of iron, 4Fe + 3O2 -> 2Fe2O3. It is caused by the metal surface acting as tiny anode and cathode areas in the presence of water containing dissolved O2 and CO2, which serves as an electrolyte, so electrons flow and the metal is eaten away.

2. Prevention of Corrosion

- Painting the surface, or coating it with grease or oil.
- Galvanizing: coating iron with a thin layer of zinc, which protects the iron by acting as a sacrificial metal — a galvanized rod of iron is coated with zinc.
- Electroplating with tin or chromium.
- Alloying the metal, e.g. making stainless steel.

3. Electroplating

Electroplating deposits a metal from its ions in solution onto the surface of an article by electrolysis. In the process of electroplating the item to be plated is made the cathode: metal ions are positively charged cations, and they are attracted to the negatively charged cathode, where they gain electrons and are deposited as a layer of metal. The anode is usually made of the pure coating metal, which dissolves to keep the concentration of ions in the bath constant. This is the principle used in tin plating and chromium plating.